Geophysicists first began to appreciate the smoldering origins of the land under the sea, known formally as ocean crust, in the early 1960s. Sonar surveys revealed that volcanoes form nearly continuous ridges that wind around the globe like seams on a baseball. Later, the same scientists strove to explain what fuels these erupting mountain ranges, called mid-ocean ridges. Basic theories suggest that because shifting tectonic plates pull the ocean floor apart along the ridges, molten rock deep within the earth's interior must rise to fill the gap. This material is produced in the second layer of the Earth's interior — the mostly solid upper mantle — and makes its way up through the crust. The collision of two plates can also result in a volcano.

5 Mark for Review

Based on the passage, why do undersea volcanoes develop?

Ⓐ	Because breaks in the ocean floor allow liquid rock from the mantle to enter.
Ⓑ	Because mid-ocean ridges pull apart when the pressure beneath them increases.
Ⓒ	Because tectonic plates accumulate along the ocean floor.
Ⓓ	Because the Earth's crust collides with the mantle.

The following text is adapted from Edith Wharton's novel *The Custom of the Country*. Mrs. Spragg has recently arrived in New York City with her daughter.

The room showed no traces of human use, and Mrs. Spragg herself wore as complete an air of detachment as if she had been a wax figure in a show-window. Her attire was fashionable enough to justify such a post, and her pale soft-cheeked face, with puffy eye-lids and drooping mouth, suggested a partially-melted wax figure which had run to double-chin.

5 Mark for Review

According to the text, what is true about Mrs. Spragg?

Ⓐ	She feels anxious in her environment.
Ⓑ	Her clothing is not suited to her position.
Ⓒ	Her features are sharp and distinctive.
Ⓓ	She appears aloof and disconnected from her surroundings.

The Importance of Being Earnest is an 1895 play by Oscar Wilde.

ALGERNON: Did you hear what I was playing, Lane?

LANE: I didn't think it polite to listen, sir.

ALGERNON: I'm sorry for that, for your sake. I don't play accurately—any one can play accurately—but I play with wonderful expression. As far as the piano is concerned, sentiment is my forte. I keep science for Life.

7 Mark for Review

Based on the text, what is true about Algernon?

Ⓐ He is embarrassed that Lane has overheard him playing the piano.

Ⓑ As a musician, he is more concerned with emotion than technical correctness.

Ⓒ Science appeals to him for the same reasons that music does.

Ⓓ He wishes that he could play the piano more accurately.

Answers: Matching Phrases

Set #1

1. C

2. E

3. A

4. B

5. D

6. G

7. F

Set #2

1. B

2. D

3. F

4. G

5. A

6. C

7. E

Answers: Literal Comprehension

1. B

The key information appears before the colon ("Interesting punctuation") in the third-to-last line. There we learn that Roosevelt's column *extended [her] reach immeasurably*, which is another way of saying that it "exposed a large new audience to [her] work." That makes (B) correct. (A) is incorrect because the passage states that "My Day" *outlasted* Roosevelt's time as first lady, but not that it "overshadowed" her achievement in that role. (C) does not fit because the passage emphasizes the wide reach of "My Day" (*six days a week in 90 newspapers*) but does not explicitly indicate that it was "among the <u>most</u> widely-read columns." (D) does not work either because the passage refers only to the *sheer frequency* of "My Day's" publication but does not make comparisons to any other column.

2. C

In this case, the answer in located in a key location, namely the last sentence. There we learn that that weirs relied on *the water's continuous motion*, or how the water naturally moves ("its existing movements"). (C) is thus correct. (A) and (B) are contradicted by the passage, which states that weirs *altered the landscape around them in distinct ways* and that they *did <u>not</u> halt the flow of the river*. Be careful with (D): the passage states only that weirs' effects were *distinct*, not that they were unpredictable.

3. A

The key phrase is in the second sentence, which states that *over time, this strategy* (i.e., reading the diary of a person in another country) *reduced cultural distance*. In other words, it "promoted perceptions of similarity," which is what (A) says. (B) incorrectly concerns citizens of the same country, whereas the findings involved citizens of different countries. Note that you can eliminate this answer for that reason even if you do not know what *antipathy* (dislike) means. (C) is incorrect because reading the diaries only caused participants to perceive people in the <u>other</u> country as more ethical—the passage does not state whether it affected their own behavior. Likewise, (D) is incorrect because the passage makes no mention of how participants' perceptions of their own cultures changed.

4. B

The answer is found in a key place, namely the end of the passage. In context of the poem's title and subject, "Youth," the statement that *Life may renew the Autumn time,/But nevermore the May* means that old age (Autumn) can stretch out for a long time, whereas youth (May) must end—i.e., "it cannot be extended." (A) is incorrect because the poem states that *swift across our dial – youth,/A* ***shifting shadow*** *goes*. (C) is incorrect because it states exactly the opposite of the correct answer: by definition, something that "cannot be extended" does not "remain perpetually (forever) fresh." Be careful with (D): the poem only states that youth is *lush with bliss*, but the question asks how youth is "unlike" old age, and the passage does not make that particular comparison.

5. A

The key information appears in the fourth and fifth sentences, which indicate that "erupting mountain ranges" (i.e., volcanoes) are believed to form when the ocean floor is pulled apart and molten (liquid) rock from the upper mantle rises into the resulting space. That corresponds directly to (A). (B) is incorrect because the ridges are pulled apart by shifting tectonic plates, not by pressure. (C) is entirely unsupported by the passage. (D) is incorrect because the passage states that volcanoes can result from "the collision of two plates," not from the collision between a plate and the mantle.

6. D

All the information you need to answer this question can be found in the first sentence: the phrases *no traces of human use* and *as complete an air of detachment as if she had been a wax figure in a show-window* correspond directly to "aloof and disconnected" in (D). (A) does not work because the passage only states that Mrs. Spragg is completely detached from her environment, not that she is "anxious." (B) is incorrect both because the passage indicates that Mrs. Spragg is dressed fashionably and because it says nothing about her actual position. (C) is directly contradicted by the passage, which indicates that Mrs. Spragg's features recall *a partially melted wax figure.*

7. B

In his short speech, Algernonon reveals that he is uninterested in playing the piano accurately ("technical correctness") and is much more concerned with sentiment ("emotion"), information that eliminates (D) and points directly to (B). (A) is incorrect because rather than being embarrassed by his presumably terrible piano-playing, Algernon seems quite proud of it. (C) is contradicted in the last two sentences, in which Algernon indicates his desire to keep music and science separate.

6 Reading for Function

If you've already spent some time studying for the SAT, you've most likely had the following experience: you see a question that asks you the primary purpose of a few lines or a passage. You go back, read the lines, and feel pretty confident that you understand what they're saying. When you look at the answers, however, they don't seem to have anything to do with what you've just read.

You go back to the passage, frantically rereading, trying to figure out what you've missed, then look back at the answers. Clear as mud. You get rid of a couple that are obviously wrong but find yourself stuck between (B) and (C), which both seem equally plausible. You remember hearing that (C) is the most common answer, so you decide to just pick it and hope for the best.

This scenario typically stems from the fact that most test-takers don't truly understand that function questions are not asking *what* the lines say but rather *why* they say it. In short, you cannot understand function without understanding content, but understanding content alone is not enough to understand function. Why? Because the SAT not only tests the ability to comprehend *what* is written in a passage but also *why* it's written. Essentially, **function questions ask you to move beyond the literal meaning of a section of a passage, or a passage as a whole, to understanding their larger purpose.**

These questions also generally require you to identify **the point that the information in question supports**. In this sense, function questions are very similar to the "illustrate" questions discussed later on, in Chapter 8: both ask you to work backwards from supporting examples to larger ideas.

However, while answers to function questions are based on the specific wording in the passage, you should keep in mind that **the answers themselves are not stated word-for-word in the text.** In fact, the answers may be phrased in much more general or abstract language than what appears in the passage; you are responsible for drawing the connection between the two. That said, you should **always keep in mind the topic of the passage because the correct answer may refer to it**, either directly or in rephrased form.

Types of Function Questions

Function questions can ask about individual sentences, or even just portions of a sentence, in which case the relevant material will be underlined. They can also ask about passages as a whole. You can expect to encounter both types of question regularly.

They are typically phrased in the following ways:

- The primary purpose of the passage is to…
- Which choice best states the function of the underlined sentence?
- The underlined portion primarily serves to…

And their answers fall into two categories:

1) Those that can **only** be answered by looking at the specific wording in the lines provided in the question. In such cases, the lines will typically contain phrasing, punctuation, or a key transition that points to a particular answer.

2) Those that **cannot** be obtained by looking at the lines provided in the question but that instead depend on contextual information.

For the second type of question especially, a sentence reference simply tells you where the information in question is located—it does not tell you the information's relationship to anything else in the passage. **The section necessary to obtain the answer will often be located either before the portion referenced in the question or, less frequently, after.**

Unfortunately, there is no way to tell upfront which category a particular question will fall into. As a result, **you should generally be prepared to read a sentence or two before and after the sentence referenced,** then focus on the appropriate lines as necessary.

Important: If the underlined section falls relatively close to the beginning of the passage, you should back up and begin reading from there. First sentences will very often give you the main point, making it much easier for you to understand the role of a particular piece of information within the passage. If the lines in questions are located in the middle of the passage—especially if the passage is on the longer side—you probably do not need to go all the way back to the beginning but can instead back up a sentence or so as necessary.

Because SAT Reading focuses heavily on relationships between ideas, it follows that questions are frequently based on the places in a passage where ideas come into contact with one another—that is, where new information is introduced, or where there is a change in focus.

The relationships between these ideas are sometimes indicated through the use of specific words, phrases, and types of punctuation, which correlate with particular function words. The chart on the next page lists some of the most common examples, along with their functions.

Functions of Key Words and Punctuation

Continuers

Continue

Additionally
Also
And
As well as
Finally
First
Furthermore
In addition
Moreover
Next
Then

Illustrate, Support

For example
For instance
One reason/another reason

Explain

Because
Explanation
That is
The answer is
The reason is

Colon
Dash

Define

That is (to say)
Properly speaking

Colon
Dash
Parentheses

Compare

As
Just as
Like(wise)
Much as
Similarly

Speculate

Could
If
It is possible
May
Maybe
Might
Perhaps

Call attention to Underscore, Highlight Emphasize

Indeed
In fact
Let me be clear

Capital letters
Exclamation point
Italics
Repetition (of a word, phrase)

Indicate Importance

Central
Crucial
Essential
Fundamental
Important
Key
Significant
The point/goal is

Draw a conclusion

As a result
Consequently
Hence
So
Thereby
Therefore
Thus

Qualify

Dashes
Parentheses

Contradictors

Contrast

(Al)though
But
Despite
However
In contrast
In spite of
Instead
Meanwhile
Nevertheless
On the contrary
On the other hand
Otherwise
Rather
Still
Whereas
While
Yet

Question, Imply skepticism

But is it really true…?

Question mark
Quotation marks

Now let's look at some examples.

Example #1

To drivers, the color red means stop, but on a map it tells traffic engineers to leap into action. Traffic control centers like the one on the seventh floor of Boston's City Hall—a room cluttered with computer terminals and live video feeds of urban intersections—represent the brain of a traffic system. <u>The city's network of sensors, cables and signals are the nerves connected to the rest of the body.</u> "Most people don't think there are eyes and ears keeping track of all this stuff," says John DeBenedictis, the center's engineering director. But in reality, engineers literally watch our every move, making subtle changes that relieve and redirect traffic.

1 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

Ⓐ To describe a problem commonly faced by traffic engineers

Ⓑ To point out some important differences between traffic control centers and the human brain

Ⓒ To discuss the purpose of items found in traffic control centers

Ⓓ To provide examples of ways in which drivers' actions can be monitored remotely

If we wanted to simplify this question, we could say something like, "Why does the passage mention *sensors, cables, and signals*?" Or, "What point is the reference to *sensors, cables, and signals* used to support?"

The line reference is smack in the middle of the paragraph, where supporting evidence usually appears. To figure out what point it supports, we must focus on the beginning and the end of the paragraph, where main points are typically stated.

The beginning of the paragraph introduces the comparison between a traffic control center and the brain. Logically, the sentence that includes the key phrase (*The city's network of sensors, cables and signals…*) must serve to further develop that comparison.

The problem here is that no answer is consistent with that idea. As a result, we must read the rest of the paragraph, paying close attention to the last sentence. The presence of the word *but* suggests that it will be very important.

What idea is presented in the last sentence? Traffic engineers are able to watch people's every move. Why? Because of the sensors, cables, and signals that relay information from the streets back to them. So the phrase in question is there to explain how traffic engineers can monitor drivers' behavior from a distance, i.e., remotely. That makes the correct answer (D).

Granted, this question isn't easy; figuring it out without consulting the answers is a challenge. At the same time, however, you cannot assume that you will automatically recognize the correct answer when you see it. Sometimes you will have to do a bit more work upfront than you'd prefer, in order to avoid confusion.

If we wanted to play process of elimination:

(A) To describe a problem commonly faced by traffic engineers

The passage doesn't discuss a problem at all. This is completely off-topic.

(B) To point out some important differences between traffic control centers and the brain

The author draws a comparison between traffic control centers and the brain, but this answer just mentions "differences," which aren't discussed at all. This answer describes exactly the opposite of what's going on in the passage. **One wrong word makes the whole answer wrong.**

(C) To discuss the purpose of items found in traffic control centers

Be careful here. The passage does mention traffic control centers and sensors, cables, and signals in very close proximity to one another; however, it states only that computer terminals and live video feeds are found in traffic control centers. In the next sentence, we learn that sensors, cables, and signals are the *nerves* present throughout the *city*. So (C) is out.

Again, that leaves (D).

If you're stuck between this option and another answer, you can follow the same steps described earlier and read to the end of the paragraph. When you get to the last sentence, you can see that the statement *engineers literally watch our every move* directly corresponds to the idea of monitoring drivers' actions remotely.

Example #2

In August 2009 a consortium of European observatories reported the discovery of COROT-7c, a second planet orbiting COROT-7. Using the data from both planets, they were able to calculate that COROT-7b has an average density about the same as Earth's. This means it is almost certainly a rocky planet made up of silicate rocks like those in Earth's crust. Not that anyone would call it Earth. <u>The planet and its star are separated by only 1.6 million miles, 23 times less than the distance between the parboiled planet Mercury and our Sun.</u> Because the planet is so close to the star, it is gravitationally locked to it in the same way the Moon is locked to Earth.

1 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

Ⓐ It emphasizes an important distinction between COROT -7b and Earth.

Ⓑ It suggests that COROT-7b could eventually come to resemble Earth.

Ⓒ It supports the hypothesis that Earth and COROT-7b may share a common origin.

Ⓓ It explains how Mercury came to occupy its position within the solar system.

Before we look at any of the answer choices, we're going to start by restating the question so that we understand exactly what it is asking:

Rephrased:

What is the purpose of the underlined statement within the passage as a whole?

OR:

What point does the underlined statement serve to support?

Next, we're going to check the lines in question and make sure we understand exactly what they're referring to:

The planet and its star are separated by only 1.6 million miles, 23 times less than the distance between the parboiled planet Mercury and our Sun.

The sentence begins with *the planet,* so we need to back up as far as necessary and determine what "that planet" is. We find the answer in the second sentence: COROT-7b. This is the noun referred to as *it* in the third and fourth sentences. That fact becomes very important because the fourth sentence gives us the purpose of the underlined sentence. So that's the part we're really interested in.

What does the fourth sentence tell us? That no one would mistake COROT-7b for Earth. In other words, it is an extremely different place. Logically, the purpose of the underlined sentence must be to expand on that idea. And that is exactly what (A) says.

One more.

Example #3

Scientists have long known that color plays a role in warning animals about danger. Only recently, however, have they begun to understand how wavelengths of light (and thus color) appear at different depths and how various marine creatures' eyes perceive this light and each other—far differently than humans see them. Where waters are murky, the majority of creatures employ nonvisual forms of communication such as smell, taste, touch, and sound. <u>But in the clear waters of coral reefs, which make up less than 1% of the world's oceans, light abounds, vision predominates, and animals drape themselves in blazing color—not only to menace potential enemies but also to evade predators, hunt for prey, and even hide in plain sight.</u>

1 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

Ⓐ	It explains how reef animals use color to hide from predators.
Ⓑ	It describes an unusual form of marine communication.
Ⓒ	It presents a novel theory about underwater perception.
Ⓓ	It emphasizes the controversy surrounding a claim about reefs animals.

In this case, we can get most of the information we need from the underlined sentence itself, although it is also helpful to look at the previous sentence.

The first thing to notice is that the underlined sentence begins with the word *but*, indicating that it includes information opposing the previous sentence or statement, and that it presents a "new idea." So it's not a bad idea to back up and get the context.

The previous sentence tells us that *the majority of [marine] creatures* employ non-visual means of communication, whereas the underlined sentence tells us that reef creatures—which, based on the 1% statistic, we can infer make up only a minuscule percentage of sea life—use vision. Essentially, the underlined sentence is describing a very unusual, or "exceptional" situation.

(B) is thus correct. The relationship is so straightforward that it is unnecessary to seriously consider any of the other answers.

Main Point vs. Primary Purpose

Thus far, we've only looked at questions asking about a specific section of a passage. However, some questions ask about the purpose of a passage as a whole. These questions straddle two categories: they are big-picture questions, but they are function questions as well and must be approached from that standpoint. Otherwise, the answer choices may not fully make sense.

Often, however, when students encounter questions asking about the primary purpose of a passage, they reiterate the main point and then become confused when an option rephrasing it does not appear.

Although the purpose and the point of the passage are related, they are not precisely the same thing, and it important to understand the difference so that you are not surprised by the wording of the answer choices.

> **Main Point –** The primary **argument** the author is making. It is usually stated more or less directly in the passage, usually in the introduction and conclusion.
>
> **Primary Purpose –** The **rhetorical goal** of the passage as a whole (e.g., *explain, emphasize, describe*). While the primary purpose is based on the overall passage, there is often a key sentence that will point to a particular answer.

In many cases, if you're clear on the main point, making the jump to the primary purpose is relatively straightforward. For instance, if the point of a passage is that Douglas Engelbart's invention of the mouse was inspired by rolling wheels called planimeters (in super-condensed SAT terms, "DE used plnmtrs → mouse"), the answer to a primary purpose question might be something as close and simple as "To describe Engelbart's invention of the mouse."

In other cases, however, the wording may change somewhat between the main point and the purpose. Continuing with the example above, the answer to a primary purpose question could be phrased in more general terms, along the lines of, "To describe the inspiration behind an invention." In that case, you would need to make the connection between the specific noun *mouse* and the more abstract words *inspiration* and *invention*.

Using the First Sentence

If a question asks about the purpose of a passage as a whole, it might seem reasonable to assume that you will need to consider, well, the whole passage. In reality, however, you may sometimes be able to identify the most likely answer based primarily on a single sentence, or even just part of a sentence. And **when it comes to purpose-of-a-passage questions, that sentence is likely to be the first one**. Just make sure you read that sentence—the whole thing—carefully, and that you skim through the rest of the passage to confirm your answer. Trust, but verify.

For example, consider the following passage.

Throughout the dinosaurs' time on Earth, there was an amplification of boniness and spikiness; <u>however</u>, the advantage of skull frills and back plates is <u>hardly self-evident</u>. The solid-domed skull of Pachycephalosaurus seems made for butting—but for butting what? The skull would be all but useless against a predator with the size and power of Tyrannosaurus Rex. The skulls of some Pachycephalosaurs, moreover, were flat and thin—a bad design for contact sports—and the spikes protruding from them were most probably blunt rather than sharp.

1 Mark for Review

Which choice best states the main purpose of the text?

Ⓐ To suggest that a longstanding hypothesis about dinosaurs may be incorrect.

Ⓑ To introduce a study and raise some questions about the validity of its findings.

Ⓒ It describes a dinosaur species in order to illustrate a puzzling trend.

Ⓓ It presents an emerging mystery and discusses an attempt to solve it.

What do we learn from the first sentence? That the *advantage of skull-frills and back plates is <u>hardly self-evident</u>*. In other words, the advantage is unclear. So logically, the purpose of the passage is to discuss that lack of clarity.

Furthermore, the presence of the word *however* in the first sentence suggests that what follows is the "new idea"—no further information will be introduced to contradict it.

The only two answers that contain words consistent with the information we've established should be present are (C) ("puzzling") and (D) ("mystery"). The passage does not provide a possible explanation ("a hypothesis") for the phenomenon in question, nor does it mention a study.

In terms of choosing between (C) and (D), look at how the answer choices are worded. If the mystery were "emerging," the passage would indicate that it was new, but there is nothing here to suggest that is the case. That eliminates (D). In contrast, (C) merely refers to "a puzzling trend." The rest of the answer is consistent with the description of Pachycephalosaurus.

For a fiction example, we're going to revisit the passage below. In Chapter 4, we looked at it in terms of the main point ("proud of family's ink business"), but now we're going to consider it in terms of its primary purpose.

In our family, the women made the ink. We stayed home. We all worked—me, GaoLing, my aunts and girl cousins, everybody. Even the babies and Great-Granny had a job of picking out stones from the dried millet we boiled for breakfast. We gathered each day in the ink-making studio. According to Great-Granny, the studio began as a grain shed that sat along the front wall of the courtyard house. Over the years, one generation of sons added brick walls and a tile roof. Another strengthened the beams and lengthened it by two pillars. The next tiled the floors and dug pits for storing the ingredients. Then other descendants made a cellar for keeping the inksticks away from the heat and cold. "And now look," **Great-Granny often bragged. "Our studio is an ink palace."**

1 Mark for Review

Which choice best states the main purpose of the text?

Ⓐ To describe the process by which a character masters a difficult task

Ⓑ To convey a character's role and sense of accomplishment in her family's work

Ⓒ To explain the influence of a distinguished relative on a character and her family

Ⓓ explore the effects of a character's decision on the members of her family

If you encounter a purpose-of-a-passage question right after you've finished reading, your first thought might be, "Oh no, I was so focused on trying to get what was going on that I wasn't really thinking about the purpose. You mean now I have to go back and reread the passage?"

In this case, if you keep the main point in mind, the primary purpose isn't much of a leap at all. In fact, the correct answer, (B), simply rewords it from a slightly different angle: the narrator takes great pride in her family's business and her role within it, and the passage serves to convey that fact. There is no need to complicate things further.

Playing Positive and Negative with Function Questions

One of the simplest ways to approach function questions and eliminate answer choices quickly is to play positive/negative. Positive passages or portions of passages typically have positive answers, while negative passages and portions of passages typically have negative ones.

Although answer choices often contain function verbs that are more neutral than the language of the passage itself, the information in the rest of the answer may be distinctly positive or negative. Even if this strategy alone does not get you all the way to the correct answer, it can allow you to quickly eliminate one or two choices upfront, giving you more time to focus on smaller distinctions between the remaining answers.

The chart on p. xx provides some examples of common positive, negative, and neutral function words that are likely to appear in answer choices.

Let's look at an example:

On what seems like a monthly basis, scientific teams announce the results of new experiments, adding to a preponderance of evidence that we've been underestimating animal minds, even those of us who have rated them fairly highly. New animal behaviors and capacities are observed in the wild, often involving tool use—or at least object manipulation—the very kinds of activity that led the distinguished zoologist Donald R. Griffin to found the field of cognitive ethology (animal thinking) in 1978: <u>octopuses piling stones in front of their hideyholes, to name one recent example; or dolphins fitting marine sponges to their beaks in order to dig for food on the seabed; or wasps using small stones to smooth the sand around their egg chambers, concealing them from predators.</u>

1 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

Ⓐ To describe ways that animals hide from predators

Ⓑ To highlight the lack of complexity in animal tools as compared to human ones

Ⓒ To provide instances of novel animal behavior in the wild

Ⓓ To emphasize the limits of animal consciousness

This is a science passage, so its tone is relatively neutral. If we look closely at the first sentence, however, the phrase *a preponderance* (very large amount) *of new evidence* implies that the author has a positive attitude toward the subject. This suggests that the correct answer will be either positive or neutral and that any negative option can be eliminated.

When we look at the answer choices, we can notice that (B) and (D) contain negative phrases (*less complex* and *limits*). Both answers can thus be eliminated immediately.

That leaves us with only two possibilities, but we still have to be careful. Remember that answers to function questions are often found **before** the underlined portions, and (A) refers to something mentioned **after**. The answer is constructed this way because many students will begin reading at the beginning of the lines referenced and overlook the information before it. Here, small stones are discussed only in relation to wasps; they are unrelated to the other animals/examples mentioned.

The point is actually found all the way back in the second sentence: *New animal behaviors and capacities are observed in the wild, often involving tool use.* In addition to the word *new*, the dashes in that sentence indicate that it is important. (C) rephrases that sentence, so it is correct.

Shortcut: (C) uses the word *novel* in its second meaning ("new"). Even in the absence of any other information, that usage suggests that (C) has an above-average chance of being correct.

Very important: While most answers contain neutral function words (e.g., *describe, emphasize*), you may occasionally encounter more extreme ones, either positive (*prove*) or negative (e.g., *condemn)*. Answers with this type of wording are usually **incorrect**. In particular, Science and Social Science passages often discuss theories that are taken seriously but have not yet been definitely (dis)proven, so answers indicating otherwise are inconsistent with how scholarly research works.

Common Function Words and Phrases

Positive	Negative	Neutral
Support Advance (a claim) Affirm Bolster Claim Defend Exemplify Illustrate Prove* Provide (evidence) Offer (an example) Substantiate **Praise** Celebrate **Acknowledge** **Propose** Imply Suggest **Emphasize** Call attention to Focus on Highlight Reinforce Reiterate Underscore **Explain** Account for Articulate Clarify Define Explicate Justify* Qualify Specify **Persuade** Advocate Encourage Promote	**Refute** Attack* Challenge Condemn* Contradict Criticize Debate Decry Deny Discredit* Dismiss Dispel Disprove* Imply skepticism Question Undermine* **Warn** Raise concern **Make fun of** Mock* Satirize Scoff at* **Concede** Acknowledge Recognize **Exaggerate** **Downplay** Minimize* **Lament*** Bemoan*	**Describe** Characterize Convey Depict Discuss Dramatize Evoke Portray Present Represent Show Trace **Indicate** Identify Point out Reveal **Introduce** **Shift** Change Digress* **Restate** Paraphrase Summarize **Hypothesize** Speculate **Analyze** Consider Describe Develop Explore Reflect on **Attribute** Cite Allude to

***Signals an answer that is likely to be incorrect.**

For a glossary of selected terms, see p. 117.

Exercise: Function

Far below the mid-ocean ridge volcanoes and their countless layers of crust-forming lava is the mantle, a 3,200-kilometer-thick layer of scorching hot rock that forms the earth's midsection and surrounds its metallic core. <u>At the planet's cool surface, upthrusted mantle rocks are dark green, but if you could see them in their rightful home, they would be glowing red- or even white-hot.</u> The top of the mantle is about 1,300 degrees Celsius, and it gets about one degree hotter with each kilometer of depth. The weight of overlying rock means the pressure also increases with depth about 1,000 atmospheres for every three kilometers.

1 Mark for Review

Which choice best states the function of the underlined sentence within the text as a whole?

Ⓐ To convey the intense pressure that pervades the mantle

Ⓑ To suggest that scientific understanding of mantle rocks is limited

Ⓒ To describe a difference between mantle rocks and other types of rock

Ⓓ To emphasize a difference between mantle rocks in different locations

The Awakening is an 1890 novel by Kate Chopin. Edna Pontellier, the protagonist, is on vacation with her husband and children at Grand Isle resort in Louisiana.

Edna Pontellier could not have told why, wishing to go to the beach with Robert, she should in the first place have declined, and in the second place have followed in obedience to one of the two contradictory impulses which impelled her. A certain light was beginning to dawn dimly within her. In short, Mrs. Pontellier was beginning to realize her position in the universe as a human being, and to recognize her relations as an individual to the world within and about her.

2 Mark for Review

Which choice best states the primary purpose of the text?

Ⓐ To highlight Edna's tendency to behave in contradictory ways

Ⓑ To present a situation in which Edna must make a momentous decision

Ⓒ To describe Edna's newfound insight into her interior and exterior existence

Ⓓ To convey Edna's sense of duty toward her family

In order to better understand people's receptiveness to opposing viewpoints, public policy scholars Julia Minson of Harvard University and Frances Chen of the University of British Columbia reviewed dozens of studies spanning 1984 to 2021. Among their findings was the fact that people who feel strongly about an issue can be receptive to others' views without altering their own opinions. As the researchers point out, two thoughtful people might examine each other's ideas seriously and, recognizing that it is possible for reasonable people to hold either perspective, respectfully agree to disagree.

3 Mark for Review

Which choice best states the function of the underlined sentence within the text as a whole?

Ⓐ It highlights a potential outcome of a strong disagreement.

Ⓑ It describes a process by which people's opinions can evolve.

Ⓒ It emphasizes the importance of considering multiple perspectives.

Ⓓ It discusses a strategy that permits opposing parties to reconcile their differences.

To be a female artist in the nineteenth century was challenging enough, but to be a female sculptor was nearly unthinkable. Not only were sculptors expected to have a familiarity with the human form that no woman in that age could acquire, but they had to work with heavy materials, such as blocks of marble weighing many hundreds of pounds. Nevertheless, a few intrepid American women wound their way to Italy and learned to sculpt.

4 Mark for Review

Which choice best describes the function of the underlined sentence in the text as a whole?

Ⓐ It discusses some of the expenses involved in becoming an artist.

Ⓑ It emphasizes the antagonism between women and the nineteenth-century artistic establishment.

Ⓒ It explains the virtual impossibility of becoming a female sculptor in the nineteenth century.

Ⓓ It describes some of the physical limitations placed on nineteenth-century women.

The following text is from Charlotte Grimké's poem "Wordsworth."

Poet of the serene and thoughtful lay!
<u>In youth's fair dawn, when the soul, still untried,</u>
<u>Longs for life's conflict, and seeks restlessly</u>
<u>Food for its cravings in the stirring songs,</u>
<u>The thrilling strains of more impassioned bards;</u>
Or, eager for fresh joys, culls with delight
The flowers that bloom in fancy's fairy realm —
We may not prize the mild and steadfast ray
That streams from thy pure soul in tranquil song

5 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

Ⓐ To criticize the human tendency toward conflict

Ⓑ To describe the power of music to express emotions

Ⓒ To highlight the consequences of uncontrolled restlessness

Ⓓ To convey the soul's youthful desire for excitement

One of the most persistent and problematic biases in science involves motivated reasoning—that is, the tendency to interpret observations to fit preconceived notions. According to Professor Brian Nosek, a specialist in human biases and co-founder of the Center for Open Science at the University of Virginia, psychologists have demonstrated that "most of our reasoning is in fact rationalization." In other words, people begin by making decisions about what to think or do, and their "explanation" later serves as a means to justify what they believed or how they intended to act in the first place.

6 Mark for Review

Which choice best states the primary purpose of the text?

Ⓐ To describe a phenomenon detrimental to the reliability of scientific findings

Ⓑ To suggest that true objectivity in science cannot be attained

Ⓒ To emphasize the inaccuracy of many scientific conclusions

Ⓓ To call attention to the dangers of motivated reasoning

In recent years, many companies have shifted from a model in which workers are placed in individual cubicles to one based on open-office plans, with the goal of fostering employee interaction and collaboration. Studies suggest, however, that such strategies may backfire, increasing job dissatisfaction and leaving workers no more likely to work together than before. Researchers at Harvard Business School theorize that employees' tendency to avoid one another in open offices may be attributable to the "fourth wall"—the imaginary curtain that prevents actors from being distracted by the audience and preserves the imaginary world of a play. To preserve a sense of psychological autonomy, employees in open offices establish their own fourth walls, which their colleagues quickly come to respect.

7 Mark for Review

Which choice best describes the function of the underlined portion in the text as a whole?

Ⓐ To describe a drawback of open offices

Ⓑ To present an explanation for an unintended phenomenon

Ⓒ To emphasize the importance of collaboration in the workplace

Ⓓ To compare office work to theatrical work

Glossary of Function Words

Account for – explain

Acknowledge (a point) – recognize the merit or validity of an idea

Advocate – synonym for *promote* and *encourage*

Bolster – support, provide additional evidence for an idea

Concede (a point) – recognize the merit or validity of an opposing idea

Discredit – disprove (literally, demonstrate a lack of credibility)

To reiterate: SAT passages are typically concerned with weighing evidence, considering prevailing theories, and proposing new explanations. Although authors and various experts cited in passages may have strong opinions about what is and is not true, passages do not, as a rule, contain sufficient information or evidence to definitively prove or disprove anything. *Discredit* is therefore unlikely to appear as a correct answer.

Dismiss – deny the importance or validity of an idea

Downplay – deliberately understate, imply that something is unimportant

Evoke – summon, call up (a memory, impression, etc.), recreate through description

Explicate – explain in great detail

Highlight – emphasize, call attention to

Minimize – deliberately understate the importance of an idea. Synonym for *downplay* and *trivialize.*

Mock – make fun of

Qualify – provide more information about a statement in order to make it seem less strong or blunt, or to indicate the conditions under which it would be true.

For example, a statement like, "The SAT is the worst test EVER" is very extreme.

To **qualify** it, however, you could say something like, "At least that's what it feels like when you're a junior in high school." That sentence reduces the impact of the first sentence, clarifies when and for whom it would be true, and makes it seem less extreme.

It can be helpful to know that qualifying phrases are sometimes **parenthetical**—that is, they are found within parentheses or dashes—and are almost like asides to the reader. For example, consider the following sentence:

> In his discovery of the law of gravity, which would transform the course of scientific thought, Newton was struck by—**if the story can be believed**—an apple that fell from a tree above the spot where he was reclining.

The phrase between the dashes is intended to suggest that this story may not in fact be true. In other words, it is intended to provide information about the questionable truth of—that is, to *qualify*—the idea that Newton was struck by an apple.

As discussed earlier, answer choices that contain familiar words used in unfamiliar ways are generally correct since the second meaning itself is being tested. Since *qualify* is not being used in its most common sense of "fulfill requirements for," it has a higher than average chance of being correct.

So here's a shortcut: If you have difficulty coming up with an answer on your own and see "qualify a statement" as an option, you should probably begin by taking a very close look at it. This is NOT to say that you should choose it without thinking, simply that you should consider it first, making sure to look back to the passage and see if it does in fact describe the function of the statement or phrase in question.

Satirize – make fun of by using irony, sarcasm, or parody

Scoff at – make fun of, suggest that something is unworthy of serious consideration

Substantiate – give evidence or support for, prove; usually too extreme to be correct

Undermine – weaken or attack the foundation of; usually too strong to be correct

Answers: Reading for Function

1. D

This is a good example of a function question whose answer depends only on the underlined sentence itself rather than on the information before or after it. The key word *but* occurs halfway through the sentence and indicates that its purpose is to describe a contrast: the first half of the sentence describes mantle rocks at the surface (green), whereas the second half describes them deep beneath the earth (very hot). In other words, it "emphasizes a difference between mantle rocks in two locations," making the answer (D). (A) is incorrect because the underlined sentence focuses on heat, not pressure. (B) is entirely off-topic, and (C) does not fit because the underlined sentence describes mantle rocks only.

2. C

The transitional phrase *In short* at the beginning of the last sentence signals that the author is about to summarize an essential idea, and indeed, the key information is located right afterwards. It involves Edna's sudden realization ("newfound insight") about herself and her position relative to the world around her ("her interior and exterior existence"). That corresponds to (C). (A) is incorrect because it takes a word from the passage (*contradictory*) and twists it beyond the scope of the passage: the narrator states only that Edna experiences contradictory impulses when Robert asks her to accompany him to the beach—we cannot infer that she has a general "tendency to behave in contradictory ways." (B) is incorrect because it is Edna's realization that is "momentous," not her decision. (D) is contradicted by the passage: the primary focus is on Edna's discovery about herself, not on her sense of obligation towards her family.

3. A

The underlined sentence begins with the phrase *As the researchers point out,* indicating that the information that follows will focus on an implication or conclusion of the research described in the previous sentences. The previous sentence discusses the finding that *two people who feel strongly about an issue can be receptive to others' views without altering their own opinions,* and the underlined sentence expands on that idea. (A) is correct because disagreeing respectfully is something that people with strongly opposing opinions might choose to do at the end of an argument—that is "a potential outcome." (B) and (D) are both contradicted by the passage. The focus here is on people whose views do <u>not</u> evolve, or who do <u>not</u> reconcile (bring together) their differences. Although the passage does discuss how individuals with different perspectives might interact, (C) is incorrect because the sentence in question does not mention "the <u>importance</u> of multiple perspectives."

4. C

The underlined sentence appears close to the start of the passage (second sentence), so start by backing up and reading from the beginning. The first sentence states that *to be a female sculptor in the nineteenth century was nearly unthinkable,* and the examples given in the underlined sentence (familiarity with human form, weight of materials) logically serve to support that claim. (A) and (B) are completely off-topic and can be eliminated. (D) refers to nineteenth-century *women,* not *female <u>artists</u>,* and is outside the scope of the passage. (C) correctly relates the idea from the first sentence that the second sentence is used to support (nearly unthinkable = virtual impossibility).

5. D

Although the underlined portion appears close to the start of the poem, in this case the first sentence is simply an exclamation that does not provide any useful information for the question. As a result, it is necessary to focus on the wording of the underlined portion, which describes the soul's search for *life's conflict, food for its craving, the thrilling strains of impassioned bards*—in other words, "excitement," which corresponds to (D). (A) is incorrect because the poem is strongly positive, whereas "criticize" is negative. (C) is fairly negative as well, and the underlined section does mention any "consequences." (B) is off-topic: while the word *songs* appears in the sentence, music is not the focus.

6. A

Although the question asks about the passage a whole, you can get a sense of the answer by using only the first sentence. (In a passage this short, the primary purpose is essentially required to be presented very close to the beginning.) The phrase *One of the most persistent and problematic biases in science* indicates that the passage will present and discuss this problem. The language is moderate and neutral, suggesting that the correct answer will be so as well. (A) is consistent with the stated purpose and does not go outside the bounds of the passage, so it is correct. (B) is too extreme—the passage only states that motivated reasoning is a problem, not that "<u>true</u> objectivity <u>cannot</u> be attained." Be careful with (C): the passage only indicates that scientists have *a tendency to interpret observations to fit preconceived notions*—it does not go so far as to emphasize that "many scientific conclusions" are actually wrong, only that they have been arrived at in a way that is not fully objective. (D) is too extreme and too negative, and the passage never explicitly discusses any dangers.

7. B

The fastest way to answer this question is to focus on the word *theorize* in the underlined portion—it indicates that the sentence in question serves to present an argument or explanation. The only answer consistent with this purpose is (B), which contains the word "explain." Otherwise, (A) is incorrect because the underlined portion does not "describe a drawback of open offices" but rather offers a theory about why a drawback (increased job dissatisfaction) exists. The beginning of the passage indicates that open offices are intended to promote collaboration but then moves to a discussion of why it does <u>not</u> take place, making (C) incorrect. Although the underlined portion does draw a comparison between open-office workers' behavior and the "fourth wall," which is an element of the theater, the main purpose of the underlined sentence is not to compare office and theatrical work but rather to explain why open offices backfire.

7 Text Completions

Every SAT contains a number of text completions: short passages that present a theory or argument and that ask you to identify the statement that logically completes the text. For example, consider the passage below. We'll look at the answer choices on the next page.

> Up close, regal angelfish flash eye-popping bands of yellow, violet, and white. But recent studies show that as regals swim against a coral reef's visually complex background, their contrasting lines merge in a predator's brain, allowing them to evade capture. According to marine biologist Gil Rosenthal, as a reef fish retreats, distance and motion can make it difficult for predators to perceive fine details and distinguish closely spaced outlines of contrasting colors. Therefore, from far away, _____

To answer this question, you must use the information presented before the blank in order infer how reef fish would be perceived by a predator far away.

Although you must make a leap of logic in order to answer a question such this, it is only a very small leap. The most important thing to understand about text completions is that they are essentially literal comprehension questions with a twist. All of the necessary information is there; you just have to put the pieces together and make explicit an idea the passage leaves unstated.

Although answers will not normally be stated word-for-word in the passage, **the text will always contain specific wording that clearly corresponds to a particular statement in the correct answer.** And very often, **the key information will appear very close to the blank**, the earlier part of the passage serving primarily to provide background for the argument.

To minimize your chances of confusion, you can follow the sequence below. You don't need to adhere to it rigidly for every question, but it provides a solid general roadmap.

1) **Carefully read the claim or theory in question.**

   Usually this will be presented right before the blank.

2) **Restate it for yourself in your own words.**

   If it is particularly dense or confusing, jot down a quick (3-6) word summary on your scratch paper. For the passage on the previous page, we could write something like "RF retreat → blend w/background." (When reef fish retreat, they blend in with their background.)

3) **Work out the implications, sticking as close to the passage as possible.**

   Given the summary above, we can conclude that *from far away,* predators have a hard time seeing brightly colored reef fish because they blend in with their background. If that seems too obvious, well… that's the point.

4) **Look for the answer that matches.**

   Keep in mind that it may use wording very different from that in the passage.

Having gone through steps #1-3, we're going to look at the answers.

Up close, regal angelfish flash eye-popping bands of yellow, violet, and white. But recent studies show that as regals swim against a coral reef's visually complex background, their contrasting lines merge in a predator's brain, allowing them to evade capture. According to marine biologist Gil Rosenthal, as a reef fish retreats, **distance and motion can make it difficult for predators to perceive fine details and distinguish closely spaced outlines of contrasting colors.** From far away, ______

1 Mark for Review

Which choice most logically completes the text?

Ⓐ the bright colors of the fish can easily be perceived, even when the water is clouded by sediment.

Ⓑ marine predators must rely on the visual aspects of their prey.

Ⓒ spots and stripes blur together, allowing even stationary fish to merge into the background.

Ⓓ the fish appear as a single mass rather than a group of individual creatures.

(A) and (B) are both directly contradicted by the passage—the whole point is that predators cannot rely on visual information to identify reef fish that they hunt. Be careful with (D): the issue is not that predators see the fish as a group, but rather that they cannot distinguish the fish from the background of the reef at all. (C), the correct answer, is directly consistent with the correct idea, which we determined beforehand: for a faraway predator, the bright patterns on the fish blend into, i.e., "merge," with the background, allowing them to escape capture.

When Is a Conclusion Valid?

When working through text completions, it can also be helpful to know something about how wrong answers are constructed. Some incorrect options will be directly contradicted by the text or be so far outside its bounds that they obviously do not fit. Identifying and eliminating these answers is a straightforward matter of reading the passage carefully and understanding the necessary vocabulary.

Other answers will be wrong in subtler ways, though. For example, they may involve **speculation**—that is, an answer *could* be true based on the information in the passage, but there isn't enough information to determine whether it is *actually* true. Half-right, half-wrong options may sound so plausible that it seems as if they should be true. (In fact, some of them may even be true in the real world.) Answers in this category can be tempting because they generally refer to claims or people that are directly mentioned in the passage.

Remember, however, that correct answers are likely to include language somewhat different from that used in the passage. It is thus crucial to not eliminate any answer without making sure you really understand what it is saying. **You must be able to distinguish between wording that rephrases important ideas from the text (valid inference) vs. wording that is genuinely off-topic (invalid).**

In addition, keep in mind that **one of the simplest ways to create a logical inference is to rewrite a statement from a different angle**—something that may involve **negative language**, e.g., the word *not* or the prefix *-un* or *-im*. For example, if a researcher claims that a star is **older** than the Earth, a correct answer might state that the star is **not younger** than the Earth.

You should also pay particularly close attention to answers that contain **double negatives**, which create a **positive meaning**. For instance, something that is "not impossible" is possible, and something that is "not unimportant" is important. This language can be very tricky, and you may need to take a few moments to work out exactly what it means.

All that said, we're going to look at a few examples of how valid conclusions can be formed.

Example #1

Researchers in Japan and Brazil have found that a particular type of muscular contraction, which occurs only when people lower weights, is most effective at increasing muscle strength and size.

There are a couple of key pieces of information here. First, the passage concerns a type of muscular contraction that is *most effective at increasing muscle strength and size.* We can therefore conclude that all other types of muscular contractions are less effective or not as effective at producing this outcome.

Next, this type of contraction occurs *only when people lower weights*, i.e., when they are not raising them.

If we put those statements together, we can logically conclude that people who want to build muscle strength and size should focus on exercises that involve lowering weights; or that raising weights is a less effective way of building muscle strength and size than lowering weights is.

Example #2

Sea turtle conservation efforts largely focus on protecting vulnerable hatchlings once they emerge. The newly hatched turtles are directed away from the bright lights of towns and encouraged to move towards the sea.

The first sentence states that conservation efforts *largely focus on protecting...hatchlings once they emerge.* We can therefore conclude that conservation efforts are less focused on protecting hatchlings before they emerge, i.e., while they are still in their shells.

Because the first sentence tells us that sea turtle hatchlings are vulnerable and need protection, we can conclude from the second sentence that *the bright lights of towns* represent danger, and that encouraging the hatchlings to move towards the sea is a way of protecting them.

We **cannot**, however, conclude that protecting sea turtles only after they hatch is the only way or the best way to conserve the species. The same is true for directing the hatchlings away from towns and toward the sea. The passage only tells us that these methods are used—it tells us nothing about their effectiveness, either in absolute terms or in comparison to other approaches.

Example #3

About 14% of major earthquakes since 2000 have been supershear events, which occur when a fault ruptures faster than seismic shear waves can travel through rock. Until recently, these earthquakes were believed to occur much less frequently because researchers had mostly looked for them on land.

We can draw several conclusions from these statements.

- If 14% of earthquakes since 2000 have been supershear events, then most earthquakes (76%) since 2000 have been other types.
- Supershear quakes occur much more frequently than researchers used to believe.
- If researchers obtained their inaccurate estimate by looking mostly at supershear quakes on land, then many supershear quakes must not occur on land. Logically, they must occur underwater instead—that is the only other option on Earth.

We **cannot**, however, draw conclusions about earthquakes—supershear or otherwise—before 2000.

Likewise, we **cannot** infer anything about why researchers focused primarily on land and neglected to consider the oceans.

Now we're going to work through a couple of additional test-style examples.

Let's start with something relatively straightforward. You don't get to see the answers just yet.

> Some people naturally exhibit a low response to training—an inability to reap the full physiological benefits of aerobic exercise. A study led by Sarah Lessard at the Joslin Diabetes Center found that participants with a low response had high levels of blood sugar—a condition known as hyperglycemia. While this condition is often associated with diabetes, it is common in non-diabetics as well. Lessard and her colleagues **predicted** that if people with hyperglycemia received a drug designed to lower blood sugar levels, they would therefore
>
> ______

We're going to begin by summarizing the scenario presented in the simplest possible terms. Do not underestimate the importance of writing things down when you are asked to juggle multiple ideas.

- People with hyperglycemia show low response to exercise.
- So: if a drug reduces hyperglycemia, **the response to exercise should get higher**.

Or, in super-condensed SAT language:

- HG ↑, exercise response ↓
- If drug makes HG ↓, exercise response ↑

The correct answer should therefore indicate Lessard + colleagues predicted that taking a drug to reduce hyperglycemia would result in an improved response to exercise.

Notice that there's nothing in there about diabetes—that information is just a distraction from the main argument. Keep that in mind when you look at the answer choices.

Ⓐ	be more likely to incorporate exercise into their daily routine.
Ⓑ	lower their risk of developing diabetes.
Ⓒ	demonstrate an improved physiological response to aerobic training.
Ⓓ	experience less severe spikes in blood sugar after consuming certain foods.

If you held onto the idea that reducing hyperglycemia would logically cause people to develop a higher response—i.e., "an improved physiological response"—to exercise, you could probably jump right to (C). The answer is so strong that it is unnecessary to consult any of the other options.

If you did want to play process of elimination, however, all the other answers are off-topic. The passage suggests nothing about lowered hyperglycemia prompting people to exercise regularly or lower their risk of developing diabetes (although the latter may be true in reality). That eliminates (A) and (B). Likewise, it says nothing about reduced hyperglycemia's influence on blood sugar after eating, eliminating (D). Again, that leaves (C) as the only option.

As a side note, the logical prediction here was actually confirmed by Lessard's study. The persistent inability of certain patients to respond normally to exercise was a mystery in the medical world for many decades, and Lessard along with her colleagues at Joslin finally solved it.

Let's look at another example—it's a bit more challenging language-wise than the previous one. We're going to use the full question from the start this time, but try to practice avoiding the answers until you've done some basic legwork.

Fully one-third of the human brain is devoted to processing visual information; in contrast, only five percent involves smell. As a result, modern neuroscience has focused most intensely on deciphering sight, with olfaction often treated as a bonus sense. That is reflected in the paucity of language to describe it, a situation that poses a problem for scientific investigation. There are countless adjectives to describe what things look—and sound—like, but humans' vocabulary for olfactory perception is fragmentary and highly inconsistent. Therefore, ______

1 Mark for Review

Which choice most logically completes the text?

Ⓐ olfactory researchers are seeking higher levels of funding to study how people process odors.

Ⓑ people often cannot explain how things smell to them in a way that is comprehensible to researchers.

Ⓒ people must be able to process visual and olfactory information simultaneously in their daily lives.

Ⓓ people should focus on sensations rather than words when they encounter new scents.

To begin, reiterate the key information from the passage and the question. Try to focus on the fundamentals and not get too distracted by challenging language. Even if you don't know what "olfaction" is, you can figure out that it refers to smelling ability: the first sentence states that only 5% of the brain involves smell, which corresponds to the idea that olfaction is treated as a *bonus sense.*

Basically, the human brain largely focuses on processing visual information and neglects scents (olfactory information). As a result, people don't have a lot of words to talk about smells. This creates a problem for studying smells scientifically.

When we summarize things this way, the question more or less answers itself. Logically, studying smells scientifically would be hard **because** people don't have a consistent vocabulary to discuss them.

If we then consider the answer choices with that information in mind, we can see that (B) is an excellent match for that idea: if people can't accurately or consistently describe what they're smelling, it becomes very difficult for researchers to gather information about how people use that sense.

Playing process of elimination, (A) is completely off-topic—the passage only states that researchers have a "problem" studying scent perception. Assuming that they are asking for more money to study it goes *way* beyond the bounds of the text.

(C) is incorrect because while people obviously do need to process both visual and olfactory (smell) information at the same time in daily life, that fact has nothing to do with the passage. It merely indicates that a much larger portion of the brain is devoted to vision and that people do not think about smell as much. This answer takes two pieces of factual information and puts them together in a way that sounds vaguely intimidating but is actually insignificant.

(D) does not fit because the passage implies no recommendations regarding what people *should* do when they encounter new scents. It only indicates that they have trouble describing smells in general.

Again, that leaves (B) as the answer.

To sum up:

Because text completions are an important part of SAT Reading, the ability to break down challenging questions of this type in a logical, step-by-step manner can have a real impact on your score.

Now, can working this carefully be a pain? Absolutely. It's much easier to just crash through questions and hope for the best. However, the bottom line is that this approach *works* (and I say this from personal experience). The SAT is a reasoning test, and these questions are explicitly constructed so that if you think carefully and reason your way through them, you can eventually find your way to the right answer.

Remember that no matter how strong your reading and/or reasoning skills are, the usefulness of taking a pencil and working out each step of the argument **by hand** on your scratch paper should not be underestimated. Again: this is not just about getting questions right but also a matter of ensuring that you do not get them wrong. After all, if you can get the points, why would you pass them up?

Exercise: Text Completions

Physicists have yet to figure out what exactly happens at the singularity of a black hole: matter is crushed, but what becomes of it then? The event horizon, by hiding the singularity, isolates this gap in our knowledge. All kinds of processes unknown to science may occur at the singularity, yet they have no effect on the outside world. Astronomers plotting the orbits of planets and stars can safely ignore the uncertainties introduced by singularities and _____

1 Mark for Review

Which choice most logically completes the text?

Ⓐ apply the standard laws of physics with confidence.

Ⓑ focus on gaining a deeper understanding of black holes.

Ⓒ attempt to peer behind the event horizon.

Ⓓ uncover phenomena not currently known to science.

Most grocery stores spray produce with water on a regular basis in order to ensure that they maintain a wholesome, fresh-picked appearance. However, according to Martin Lindstrom, author of *Brandwashed: Tricks Companies Use to Manipulate Our Minds and Persuade Us to Buy*, not only does this liquid lack any practical purpose, but it actually has a deleterious effect: _____

2 Mark for Review

Which choice most logically completes the text?

Ⓐ shoppers are unlikely to purchase fruits and vegetables that appear dry and withered.

Ⓑ moisture causes picked vegetables to spoil more quickly than they otherwise would.

Ⓒ certain vegetables lose some of their nutrients when they are boiled.

Ⓓ produce must be watered at predictable intervals in order to appeal to consumers.

Although it is widely assumed that cognitive bias clouds our assessment of the people around us, their research and that of others, a group of researchers at the Santa Fe Institute has found that people's estimations of what their friends and family believe are often largely correct. That's because as highly social creatures, we have become very good at sizing up those around us—what researchers call "social sensing." It is therefore possible ______

3 Mark for Review

Which choice most logically completes the text?

Ⓐ to gather highly accurate information about trends by asking about individuals about their social circles rather than their own beliefs.

Ⓑ to determine people's views on a variety of topics by analyzing the ways in which they interact with others.

Ⓒ to discover what people truly believe about an issue by asking them to reflect on their personal biases.

Ⓓ to develop an algorithm that reliably predicts people's preferences about a wide range of items.

One of the most startling discoveries of the early 21st century was that Indo-European languages seem not to have been spread by Anatolian farmers living in what is now Turkey, as was commonly thought, but rather by a people called the Yamnaya, horse-herding nomads who lived on the Eurasian steppes more than 5,000 years ago. A host of linguistic evidence suggesting this possibility was first compiled persuasively by archaeologist David Anthony in 2007; DNA evidence later proved he was on target, showing that ______

4 Mark for Review

Which choice most logically completes the text?

Ⓐ members of tribes from the steppes arrived in Germany sometime between 2500 and 2000 BCE.

Ⓑ the Yamnaya were a genetic blend of three separate Eurasian populations.

Ⓒ around 5,000 years ago, the Yamnaya's genes began to appear throughout Europe and Asia.

Ⓓ the Yamnayans were linguistically unique in comparison to other groups from the same period.

When Isaac Newton published the *Principia* in 1687, his laws of motion solved numerous problems in physics; however, they also introduced a new conundrum, which was not fully grasped until centuries after Newton and which still poses a problem for cosmologists today. Essentially, Newton's laws work about twice as well as they are intended: they describe the everyday world that people move through, but they also account perfectly well for a world in which people walk backwards, clocks tick from evening to morning, and ______

5 Mark for Review

Which choice most logically completes the text?

Ⓐ objects interact unpredictably with one another.

Ⓑ planets that are in motion remain in motion.

Ⓒ particles of different weights move at varying speeds.

Ⓓ apples rise from the ground to the branches of a tree.

Exactly how Mars was formed approximately 4.5 billion years ago is a mystery, although there are several theories. One idea is that the planet was created via a titanic collision of rocks in space, spawning an all-encompassing magma ocean. When it cooled, a crust with high levels of basalt was formed. Another possibility is that parts of the first crust on Mars had a different origin, one that would primarily show large concentrations of silica. Planetary geochemist Valerie Payré and her partners analyzed data for the planet's southern hemisphere, the planet's oldest region. They discovered nine locations rich in feldspar, a mineral associated with lava flows that are higher in silica than basalt. This finding led them to conclude that ______

6 Mark for Review

Which choice most logically completes the text?

Ⓐ portions of Mars' surface were never covered by a crust.

Ⓑ the magma ocean formed from rocks colliding in space was not all-encompassing.

Ⓒ the southern hemisphere of Mars contained more silica than was previously believed.

Ⓓ the first crust on Mars did not develop until long after the planet was formed.

Answers: Text Completions

1. A

The second-to-last sentence includes the transition *yet*, which indicates key information: although singularities may involve all sorts of mysterious processes, they have no real-world effect. Thus, scientists can ignore them. The correct answer must restate or be consistent with this idea. (A) is correct because if singularities don't have an effect, then scientists can treat everything as normal and apply the usual laws of physics. (B) is unrelated to the required idea, and (C) and (D) both contradict the idea of ignoring puzzling aspects of black holes and maintaining the existing scientific framework.

2. B

The key phrase occurs right before the blank: *not only does the liquid [used to water fresh produce] lack any practical purpose, but it also has a deleterious* (negative] *effect*. The information that follows must logically explain that effect. (A) does not explain the negative impact of watering on already-picked fruits and vegetables—the fact that shoppers are unlikely to select produce in poor condition is irrelevant to this section of the passage. (C) is incorrect because the passage focuses only on the effects of water on uncooked vegetables. (D) incorrectly states a positive effect of watering produce. (B) is correct because faster spoilage is a negative effect of spraying fresh fruits and vegetables with water.

3. A

This is a good example of a question in which the answer is very nearly stated in the passage. The text makes clear that people are generally pretty accurate in identifying the beliefs of their friends and family members, so logically it must be possible to gather accurate information about trends by asking individuals what the people close to them think rather than what they themselves think—which is exactly what (A) says. (B) is incorrect because the passage focuses on beliefs, not interactions. (C) says exactly the opposite of what is indicated by the passage—reliable information can be obtained by asking people about others, not themselves. (D) goes way beyond the scope of the passage, which does not discuss the role of technology in determining people's preferences at all.

4. C

The statement immediately before the blank indicates that David Anthony was *on target*, so the blank must refer to evidence supporting his theory. What was Anthony's theory? That Indo-European languages were spread by the Yamnaya, *horse-herding nomads who lived on the Eurasian steppes more than 5,000 years ago*. Note that to identify this information, you must essentially back up to the beginning of the passage—the phrase *this possibility* in the last (second) sentence refers to an idea described in the previous (first) sentence, and there is no way to make sense out of the passage without knowing what that idea is. The correct answer must support it with information indicating that the Yamnaya moved throughout the regions, plural, where Indo-European languages became established. (A) focuses on Germany, which is only one area, so it can be eliminated. (B) is off-topic—the Yamnaya's genetic roots are irrelevant to the argument. (D) does not fit because the correct answer must involve the Yamnaya's migration and its eventual effects. Even though this option mentions the Yamnaya's language, it does not mention how its influence was spread. Only (C) contains that information: if "the Yamnaya's genes appeared throughout Europe and Asia," these people must have been present in many regions, hence the spread of their language.

5. D

The first two examples in the list describe situations in which things run backwards, so logically, the final item in the list must describe a normal occurrence that is reversed. Objects interacting unpredictably; planets remaining in motion; and particles of different weights moving at varying speeds are not examples of a phenomenon running backwards. Only apples rising back to a tree, rather than falling from one, illustrates the required idea. (D) is thus correct.

6. B

This is a complicated question, so we're going to break it down into parts. Notice that the process outlined here requires you to work backwards through the passage. First, what is the finding? That nine regions of Mars were rich in feldspar, which is *associated with lava flows that are higher in silica than basalt*. What is the significance of this finding? It supports the theory that <u>*parts*</u> *of the first crust on Mars had a* <u>*different*</u> *origin*. A different origin from what? From the <u>*all-encompassing*</u> *magma ocean* that was created from rocks colliding in space. Logically, if parts of the crust were formed from something other than the magma ocean, then the ocean could <u>not</u> have been *all-encompassing* (covering everything). And that is what (B) says. (A) is incorrect because the passage only discusses the origins of the first crust that did exist on Mars—there is nothing to suggest that there were places without a crust. (C) is incorrect because the passage indicates only that feldspar, which is found in Mars's southern hemisphere, is *higher in silica than basalt*. The passage does not indicate how much silica researchers previously believed was present. (D) does not work either because the passage only mentions that Mars was formed around 4.5 billion years ago and the southern hemisphere is Mars's oldest region—there is no mention of when the crust formed, and the amount of feldspar and silica in that area is not discussed in relation to any particular timeline.

8 Supporting & Undermining

"Support" and "undermine" questions belong to the same inference family as text completions but require you to go a step further. Rather than just ask you to identify a logical conclusion, they ask you to move beyond a stated hypothesis or conclusion and identify a statement that would either support or undermine (weaken) it.

While some simpler questions asking test-takers to support or illustrate an argument may appear at the beginning of a Reading/Writing section, the most challenging of these items are likely to appear among the later Reading questions—that is, just before the first Writing questions.

As is true for text completions, if you approach these questions methodically, they have the potential to become quite straightforward. **But you can't get impatient, and you can't skip steps.** If you're not really certain what a question is asking, OR you don't feel that you can focus properly, you are better off skipping it and returning to it after you have answered the Writing questions.

The process for answering support/undermine questions can be broken into three main steps:

1) Identify the claim and rephrase it if necessary.

If the claim or theory is stated simply in the question, take a second and process it so that you're clear on what it is. If it worded more complexly, rephrase it in simpler language and jot it down on your scratch paper. You can't determine whether a set of lines would support or weaken an idea unless you know what that idea is.

2) Determine what sort of information would support or weaken the claim.

You should at least attempt to get a basic sense of this on your own. Do not assume you'll be able to recognize the information from the answer choices, which by definition are written to confuse you.

3) Check the answers.

If you've done steps one and two carefully, the option closest to what you've said should be correct.

Illustrating a Claim

We're going to start with the most straightforward type of "support" question, which asks you to identify the statement that best illustrates a main idea.

In comparison to the question types we'll look at later in this chapter, "illustrate" questions are more likely to accompany **prose fiction** or **poetry** passages. As always, you are not expected to have any pre-existing knowledge of the work in question—all the necessary background information will be explicitly provided.

That said, you may still need to navigate challenging, old-fashioned, and possibly metaphorical language in order to understand what an author or poet is literally saying. For example, consider the following question.

"Slow Through the Dark" is an early 1900s poem by Paul Laurence Dunbar. In the poem, the speaker urges the listener not to lose hope, even in difficult circumstances.

1 Mark for Review

Which quotation from "Slow Through the Dark" most effectively illustrates the claim?

Ⓐ "Slow moves the pageant of a climbing race; / Their footsteps drag far, far below the height,"

Ⓑ "No strange, swift-sprung exception we; we trace / A devious way thro' dim, uncertain light,— ."

Ⓒ "Heed not the darkness round you, dull and deep; / The clouds grow thickest when the summit's nigh."

Ⓓ "Who stoppeth here to spend a while in sleep / Or curseth that the storm obscures the sky?"

Let's start by reiterating the basics: the question is asking us to identify the lines in which the speaker <u>urges</u> the listener to remain <u>hopeful</u> in a tough situation. As a result, the correct answer must convey the idea of hope.

We can assume upfront that (D) is incorrect because it cites a question, which goes against the idea of urging (encouraging) someone. (A) does not fit either because its focus is on [dragging] footsteps, which has nothing to do with encouraging someone to be hopeful. (B) can likewise be eliminated: the word *we* includes the speaker, and we want an answer in which the speaker addresses someone else.

(C) is the best fit. The phrase *Heed not the darkness round you*—in other words, do not give into despair— is essentially an order not to lose hope when circumstances are difficult.

Supporting a Claim

"Support" questions are similar to "illustrate" ones, but they are more likely to accompany science or social science passages and involve logical arguments rather than works of literature.

For example, take a look at the question below.

Marketers assume that the more choices they offer, the more likely customers will be able to find just the right thing. They assume, for instance, that offering 50 styles of jeans instead of two increases the chances that shoppers will come across a pair they really like. <u>Nevertheless, research now shows that there can be too much choice; when there is, consumers are less likely to buy anything at all, and if they do buy, they are less satisfied with their selection.</u>

1 Mark for Review

Which finding, if true, would most strongly support the conclusions of the research described in the underlined section?

Ⓐ People faced with choosing among a large number of options experience physical symptoms associated with feeling overwhelmed.

Ⓑ Consumers given a choice of 10 jams in a taste test were more likely to make a purchase than consumers who were offered only five.

Ⓒ Shoppers who purchased food from multiple stores spent more on groceries than shoppers who purchased all their food from a single store.

Ⓓ Some companies have increased the number of products they offer in order to appeal to a wider range of consumer preferences.

As always, we're going to start by summarizing the relevant section of the passage, in this case the research described in the underlined section.

What does that research show? That when faced with too many choices, consumers are less likely to make a purchase, and if they do make one, they are less likely to be happy with it. The correct answer must include a finding consistent with that idea.

(B) can easily be eliminated since it directly contradicts the passage—people given more choices would be less likely to buy jam, not more. (C) and (D) go in the wrong direction as well since they imply that more choice leads to a positive outcome.

Only (A) fits. Feeling overwhelmed when confronted with too many options would make people less likely to make a choice or be happy with a choice they've made. (A) is thus correct.

Let's look at another, more challenging question.

As a person sleeps, the motor cortex—the part of the brain that controls movements—replays skills that it learned during the day. In a recent experiment run by the BrainGate consortium, researchers observed that the pattern of firing neurons sped up during sleep, echoing findings from previous animal studies. The scientists were also interested to note that replay took place not during REM sleep, which is when people normally dream, but during deep, slow-wave sleep, which occurs in the first three or four hours of the night. They theorize that replay during this period helps the brain consolidate new information, moving it from short-term to long-term memory.

1 Mark for Review

Which finding, if true, would most strongly support the scientists' theory?

Ⓐ Athletes who slept for at least seven hours before a game competed at a higher level than ones who slept for less time.

Ⓑ People whose neuron firings increased in speed during REM sleep retained new information over a longer period of time than people whose neurons fired more quickly early in their sleep cycle.

Ⓒ Students who slept for only a few hours before an exam showed no difference in performance when compared to students who slept for a full night.

Ⓓ Dancers who slept for several hours shortly after learning a new routine knew it better two weeks later than a group that stayed awake.

The question indicates that we will need to identify a statement that supports the scientists' theory, so it is necessary to read the passage with that key word in mind. It appears in a different form in the last sentence, telling us scientists ***theorize*** *that replay during the period* (i.e., during slow-wave sleep, early in the night) *helps the brain consolidate new information, moving it from short-term to long-term memory.*

Next, we need to think about what type of scenario would support the claim that replaying new skills during slow-wave sleep helps people retain what they have learned in the long term.

(A) No. This answer only involves an amount of sleep; it says nothing about consolidating new skills.

(B) No, but be careful. This choice refers to "retaining new information" but incorrectly ties it to REM sleep. The passage, in contrast, discusses a phenomenon that does NOT occur during REM sleep.

(C) No. The passage discusses an effect that <u>improves</u> learning – if two groups performed identically, that would not be the case.

(D) Yes, this fits. The passage discusses a phenomenon that happens early in the sleep cycle, which means that it <u>would</u> occur in people who only slept for several hours. If dancers who slept that amount after learning a new routine knew it better after two weeks (i.e., longer-term) than dancers that did not sleep at all, that supports the idea that replay early in sleep consolidates learning.

Undermining a Claim

"Undermine" or "weaken" questions are based on the same logic and involve the same process as "support" questions – they just go in the opposite direction, asking you to identify the statement most inconsistent with a given theory or conclusion.

For example, consider the passage below.

Dinosaurs, with the exception of the ancestors of birds, disappeared during a mass extinction 65 million years ago when an asteroid struck the Earth. Because a high metabolic rate has generally been suggested as one of the key advantages when it comes to surviving mass extinctions, some genetic paloeontologists have proposed that birds survived while non-avian dinosaurs did not because of the birds' increased metabolic capacity.

1 Mark for Review

Which finding, if true, would undermine the genetic paleontologists' proposal?

Ⓐ The metabolisms of non-avian dinosaurs may have decreased over time.

Ⓑ Because some non-avian dinosaurs moved quickly, they likely had high metabolic rates.

Ⓒ Birds have some of the highest metabolism rates of any creatures on Earth.

Ⓓ Many dinosaurs with very high metabolic rates went extinct 65 million years ago.

The first step is to identify the genetic paleontologists' proposal – without that information, we cannot know what sort of statement would weaken it. The information appears in the last sentence: *birds survived [the mass extinction 65 million years ago] while non-avian dinosaurs did not because of the birds' increased metabolic capacity.*

In other words, metabolic capacity ↑, chances of surviving a mass extinction ↑.

To weaken that claim, we need to find an answer supporting the opposite:

- Metabolic capacity ↑, chances of surviving a mass extinction ↓.

OR:

- Metabolic capacity ↑, no effect on chances of surviving a mass extinction.

If we hold very tight to that idea and don't allow ourselves to become confused by any of the other options, we can identify (D) as the sole choice consistent with that idea. By definition, the extinction of creatures with high metabolic rates contradicts the claim that having a high metabolic rate prevents animals from going extinct.

Note that not trying too hard to wrap your head around the wrong answers is really key here. If you got caught up in trying to work out the logic behind them, you could really waste a lot of time – which is itself a sign that those options are incorrect.

Exercise: Supporting and Undermining

"Gerarda" is an 1895 poem by Eloise Bibb. In the poem, Bibb emphasizes the contrast between the way in which objects are depicted in Gerarda's paintings and their true appearance, writing ______

1 Mark for Review

Which quotation from "Gerarda" effectively illustrates the claim?

Ⓐ But in her spiritual world she leaves
Her mind, her thoughts, her soul, her brain,

Ⓑ Her paintings hang upon the wall,
The power of genius stamps them all;

Ⓒ Now to-day o'er canvas bent,
She strives to place these visions sent

Ⓓ And thus her pictures plainly show,
Not nature's self but ideal glow.

"Frederick Douglass" is an 1895 poem by Paul Laurence Dunbar. In the poem, Dunbar praises Douglass for his honesty and refusal to be intimidated, writing ______

2 Mark for Review

Which quotation from "Frederick Douglass" effectively illustrates the claim?

Ⓐ No miser in the good he held was he,--
His kindness followed his horizon's rim

Ⓑ A hush is over all the teeming lists,
And there is pause, a breath-space in the strife;

Ⓒ And he was no soft-tongued apologist;
He spoke straightforward, fearlessly uncowed;

Ⓓ He was her champion thro' direful years,
And held her weal all other ends above.

"Left-digit" bias, in which people focus on the first digit of a number and ignore the last, is a well-documented phenomenon that explains why prices tend to end in "9". According to a recent study by researchers in California and Sweden, it also influences journalists' coverage of unemployment rates. Even when jobless levels and the amount of change in unemployment rates are similar in two regions, crossing a round number informs reporters' assessment of the story's newsworthiness—even when both figures fall within the same margin of error.

3 Mark for Review

Which finding, if true, would most directly support the researchers' finding?

Ⓐ	The Bureau of Labor Statistics publishes a report of unemployment rates by state, rounding its statistics to the nearest decimal place.

Ⓑ	The mention of high unemployment figures in a newspaper caused consumer spending to drop by almost 2.5% in one city.

Ⓒ	A city newspaper ran several stories about joblessness when the unemployment rate reached 5% but quickly stopped coverage of the issue.

Ⓓ	A newspaper published significantly fewer stories about local unemployment when the jobless rate was 3.9% than when it rose to 4.1%.

One argument that is commonly cited to support the idea that birds lack a sense of smell is that some birds' olfactory bulbs are relatively small. As a result, many scientists concluded that these creatures gave up smell in favor of improved eyesight. This notion became so pervasive that it once was repeated to avian expert Danielle Whittaker as fact by a prominent neurobiologist.

4 Mark for Review

Which finding, if true, would most undermine the neurobiologist's statement?

Ⓐ	An analysis of avian genomes revealed the presence of proteins that bind to odors and relay a signal to the brain.

Ⓑ	The number of olfactory neurons is much smaller in birds than in most other animals.

Ⓒ	The odors that birds depend on for food and social relationships are quickly dispersed in the wind.

Ⓓ	Birds sometimes overlook strongly scented prey in favor of animals that are more visually striking.

New research suggests that coffee could have a positive effect on cardiovascular health. Although caffeine is coffee's most well-known constituent, the beverage contains more than 100 biologically active components. In a study led by Jiyoung Kim, researchers at Seoul National University concluded that non-caffeinated compounds likely play a role in the positive relationship between coffee consumption and health.

5 Mark for Review

Which finding, if true, would most directly support the researchers' conclusion?

Ⓐ	Whereas regular coffee contains from 70-140 mg of caffeine per cup, decaf contains approximately 0-7mg
Ⓑ	Decaffeinated coffee causes cells to produce NQO1, an enzyme that has neuroprotective benefits.
Ⓒ	Extracts from caffeinated coffee have been shown to aid weight loss more than a placebo.
Ⓓ	People who drink several cups of coffee every day are less vulnerable to certain diseases.

Scientists have long known that self-pollination can inhibit plants' ability to adapt to environmental changes, but until recently they did not know exactly how or how quickly the changes occurred. A group of researchers at Washington State University set up a controlled greenhouse experiment in which a group of monkeyflower plants were isolated from the bumblebees that normally pollinate them. Initially, the plants produced few seeds, but as they adapted to the self-pollination process, their seed production increased dramatically. In addition, their flowers changed shape to facilitate the transfer of pollen. However, scientists expected that the plants would eventually become more vulnerable to shifts in their environment.

6 Mark for Review

Which finding, if true, would most directly support the scientists' expectation?

Ⓐ	Some flowers also received pollen from nearby flowering plants, a process that is known as geitonogamy.
Ⓑ	The plants' genetic variation decreased by 24 % over nine generations, making them more susceptible to a variety of pathogens.
Ⓒ	None of the plants contained the types of genetic defects that are typically found in the wild.
Ⓓ	By eliminating the transfer of pollen grains, the plants were able to reduce the amount of pollen wasted by 60%.

It was only when Alma Thomas turned away from figurative art and toward abstraction that she rose to acclaim as an artist. When the solidity of her line gave way to broken, vibrant colors, the beauty she had long seen emerged.

7 Mark for Review

Which choice, if true, would most effectively support the writer's claim?

Ⓐ Her painting *Starry Night and the Astronauts* (1972) includes a small kaleidoscope of red, orange, and yellow that suggests the spaceship Apollo 10.

Ⓑ Thomas was fascinated by the natural world, incorporating everything from the flowers in her garden to the stars in the night sky into her paintings.

Ⓒ Her *Still Life with Vases and Flowers* (1964), which contains realistic images, feels labored, whereas *Lunar Surface* (1970) glows with rich splashes of purple and blue.

Ⓓ Although Thomas's own works focused on nature, she believed that art could also evoke the energy of airplanes, cars, skyscrapers, and electric signals.

Middlemarch is an 1871 novel by George Elliot. In the story, a young woman named Dorothea Brooke becomes engaged to a scholar named Mr. Casaubon. In describing Dorothea's motives for marriage, Elliot emphasizes her belief that she and Mr. Causabon have a great deal in common.

8 Mark for Review

Which quotation from *Middlemarch* effectively illustrates the claim?

Ⓐ Dorothea by this time had looked deep into the ungauged reservoir of Mr. Casaubon's mind, seeing reflected there in vague labyrinthine extension every quality she herself brought;

Ⓑ If it had really occurred to Mr. Casaubon to think of Miss Brooke as a suitable wife for him, the reasons that might induce her to accept him were already planted in her mind.

Ⓒ His notes already made a formidable range of volumes, but the crowning task would be to condense these voluminous still-accumulating results and bring them to fit a little shelf.

Ⓓ But in this case Mr. Casaubon's confidence was not likely to be falsified, for Dorothea heard and retained what he said with the eager interest of a fresh young nature to which every variety in experience is an epoch.

Answers: Supporting and Undermining

1. D

The question indicates that the correct answer must support the idea that Bibb "emphasizes the contrast between appearance and reality." As you read through the answer choices, look for a word such as *but* or *yet* that will indicate a contrast: (A) and (D) both contain *but*, whereas (B) and (C) do not include this type of language and can be assumed to be incorrect. (A) does not fit because it does not mention Gerarda's paintings, whereas (D) both mentions her "pictures" and refers to "not nature's self (= reality) but ideal glow (= appearance)." (D) is thus correct.

2. C

Start by defining the claim: *Dunbar praises Douglass for his honesty and refusal to be intimidated.* The correct answer must be positive and illustrate these qualities. (A) refers to "kindness," but that is not quite a match, so this option can be eliminated. (B) is a relatively neutral description that does not involve the necessary criteria. Although (D) is positive, this answer is off-topic as well. (C) is correct because it indicates that Douglass felt no need to apologize, spoke in a "straightforward" manner, and was "fearlessly uncowed" (intimidated).

3. D

If you scan the passage, you'll find the word *researchers* in the second sentence. The rest of the passage is essentially dedicated to describing what they found: "left-digit" bias affects whether journalists cover unemployment rates, and employment figures that cross a round number (a number ending in zero) attract more attention than ones that do not. The correct answer must therefore provide an example of a situation in which coverage of a jobless rate increased when it crossed a round number. The only answer that meets this requirement is (D): the change from 3.9% to 4.1% unemployment was very small, but the jump over a round number correlated with greater coverage of the issue. (A) is completely off-topic, and (B) does not fit because the passage does not even mention consumer spending. (C) describes an increase in unemployment coverage when the jobless rate reached a round number (5%), but the last part of the answer ("quickly stopped coverage") does not fit. There is also no comparison to the amount of coverage unemployment received when the rate was below 5%, making this a weaker answer than (D).

4. A

What did the neurobiologist say? That *these creatures* (i.e., birds) *gave up smell in favor of improved eyesight.* This is an "undermine" question, so the correct answer must contradict that idea. Most logically, it will indicate that birds have both good eyesight and a sense of smell. The relationship between (A) and the claim in question may not be immediately obvious, but consider it carefully: if avian (bird) genomes suggest that birds are physiologically capable of perceiving odors, that would suggest they do in fact have a sense of smell. So keep (A). The presence of an unusually small number of olfactory neurons in birds would support rather than undermine the claim in question, so (B) can be eliminated. (C) and (D) have the same problem. In (C), if birds cannot rely on odors to find food, that would support the idea that they did not evolve a sense of smell. And in (D), ignoring strong scents in favor of visual information suggests a poor sense of smell.

5. B

What is the researchers' conclusion? That *non-caffeinated compounds likely play a role in the positive relationship between coffee consumption and health.* The correct answer must therefore be consistent with the idea that something other than caffeine is responsible for coffee's health benefits. (A) has no relationship to this argument; the fact that decaf contains trace amounts of caffeine is irrelevant. (C) is incorrect because it focuses on the health benefits of caffeinated rather than decaffeinated coffee, and (D) does not quite work because it does not indicate what type of coffee is involved. It supports the idea that coffee has health benefits, but not the idea that something other than caffeine is responsible for them. Only (B) works: if decaf causes the production of an enzyme with "neuroprotective benefits" (i.e., that protects the brain), then caffeine is presumably not the cause.

6. B

What is the scientists' expectation? If you're not sure where to look for this information, focus on the end of the passage, where key information in support/undermine questions (and text completions) is most likely to be located. Indeed, the word *expected* appears in the last sentence. Basically, the scientists expected that although the self-pollinating plants appeared to be doing well, they would eventually develop a weakness that would make them *more vulnerable to shifts in their environment.* The correct answer must be consistent with that idea. (A) and (D) are off-topic and can be eliminated. (C) would weaken rather than support the expectation: plants without genetic defects would be <u>less</u> likely to become vulnerable to disease. (B) is correct because an almost 25% decrease in genetic diversity that left plants more "susceptible" (i.e., *vulnerable*) to diseases would be directly consistent with the researchers' expectations.

7. C

What is the writer's claim? That the beauty of Alma Thomas's work emerged when she moved away from figurative art and began to work abstractly. The correct answer must support or illustrate that idea. (A) mentions Thomas's use of color but does not indicate whether this painting was figurative or abstract. (B) is off-topic, focusing on the subject matter of Thomas's work rather than its style. (D) is incorrect for the same reason. Only (C) offers an explicit contrast between Thomas's "labored" early figurative work and "glow[ing]" abstract later work.

8. A

What is the claim? That Dorothea believes she and Mr. Casaubon have many things in common. The correct answer must therefore emphasize Dorothea's belief in their similarity. The main challenge in this question is sorting through the dense, old-fashioned language and understanding what each quotation is literally saying. (A) is correct because it indicates that Dorothea sees "every quality" of her own reflected in Mr. Casaubon. In other words, she perceives him as very similar, if not downright identical, to her. (B) only indicates that Dorothea ("Miss Brooke") is already interested in marrying Mr. Casaubon—it says nothing about their similarities. (C) is off-topic, focusing on Mr. Casaubon and not mentioning Dorothea at all. (D) is incorrect as well because it only conveys the idea that Dorothea is extremely interested in what Mr. Casaubon has to say because of her youth; it does not explicitly mention that she views him as being similar to her.

9 Graphs and Charts

Graph- and chart-based questions come in a variety of shapes, sizes, and layouts, and range from simple and straightforward to seemingly very complex. While they tend to involve scientific topics, you are likely to encounter some social science ones as well. They may also overlap with other question types, for example doubling as text completions or support/undermine questions. If you do well on Reading/Writing Module 1, you should expect to encounter one or two more challenging items in Module 2.

In general, the information most directly relevant to the question will be presented at the **end** of the passage. As is true for text completions, the first sentence or two will often serve primarily to provide context. In many if not most cases, you can therefore skim through the introductory material and then slow down towards the end, paying very close attention to the last sentence. If you find it helpful to do so, you may even want to highlight the key words so that you know exactly what type of information you are looking for.

In principle, graph questions are designed to make you synthesize multiple sources—a passage, a question plus answer choices, and a graphic. In reality, however, these items are rarely as complicated as they appear.

The first thing to keep in mind is that you are not expected to have any prior knowledge of the topics. No matter how complex the terminology may seem—and you are likely to encounter some very long and technical terms—all of the information you need to answer the questions will be right there in front of you. Provided that you get the gist of the key facts, you do not need to understand every word to get questions right.

The second thing to understand is that **while some questions may genuinely require you to use the passage along with information from the graph or chart, many answers can be identified based strictly on the wording of the question and the answer choices—it is unnecessary to look at the graphic at all**. This is particularly true for very complex-looking questions, which are essentially constructed to make you think that they are much harder than they actually are.

Even if information from the graph or chart is required to answer the question, you should **always begin by carefully reading the passage — usually the last sentence — and the question because these places tell you what theory, claim, or conclusion to focus on**. Only a portion of the information in the graph will be relevant, and if you don't know where to direct your attention, you will waste both time and energy looking for... something or other. What has the potential to be a simple matter of "same idea, different words" can morph into something unnecessarily complex and confusing.

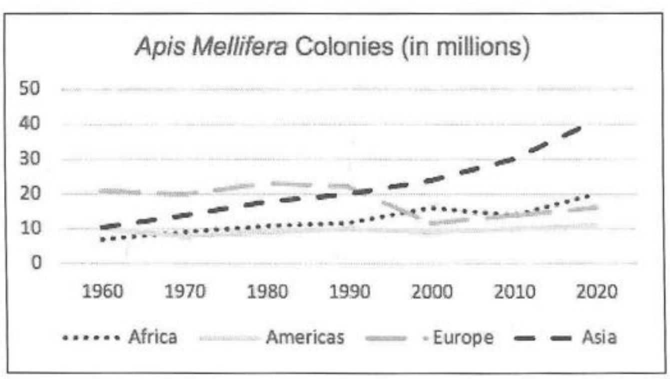
In some instances, multiple answer choices will cite accurate information from the graphic; however, only one option will correspond to the specific focus in the question. Thus, **it is entirely possible to interpret a graph or chart perfectly and still end up with the wrong answer**. In such cases, ignoring the chart and focusing on the phrasing in the text portion of the question may, paradoxically, give you a much better chance of getting the question right.

While it is not a bad idea to double-check your answer against the graph, just to make sure you haven't overlooked anything, be careful not to start second-guessing yourself. If only one option matches the specific criteria indicated in the question, then it must be the correct answer by default.

Starting below, we're going to look at some examples of how to work through graphic-based questions, both when the graph is necessary and when it isn't.

Example #1

Let's start with something reasonably straightforward.



As part of a unit on pollination, a biology student is researching Colony Collapse Disorder, the mysterious disappearance of millions of honeybee (*Apis mellifera*) colonies in the early 2000s. Although many wild bee populations have continued to decline, the number of honeybees has actually increased on every continent. The student wants to **emphasize that in some regions, honeybee populations had not only rebounded by 2020 but also reached their highest level in 60 years.**

1 Mark for Review

Which region included in the graph should the student cite to support this claim?

Ⓐ Africa

Ⓑ Americas

Ⓒ Europe

Ⓓ Asia

As discussed earlier, we're going to start by focusing on the passage, particularly the last sentence. The word *emphasize* tells us that the key information will follow.

What does the "student" want to emphasize about honeybee populations in some regions in 2020? The sentence tells us that there are two conditions that must be met:

1) They had rebounded (i.e., increased after a decline).

2) They had reached their highest level since 1960 (the beginning of the graph).

For an answer to be correct, **both** of these things must be true.

In this case, the question is constructed so that we have no choice but to look at the graphic—the answer choices themselves provide no clue as to which option might be correct.

The graph itself does not contain any tricks: the four lines correspond to rises and falls in honeybee (*Apis mellifera*) populations on four continents during a 60-year period (1960-2020).

- To meet the "rebound" requirement, the correct line must go down and then up.
- To meet the "highest point" requirement, the line must be higher for the 2020 point than for any other year since 1960.

The only line that fits both those criteria is the line for Africa: there is a dip between 2000 and 2010, and the highest point (20 million) corresponds to 2020. (A) is thus correct.

Although the changes in the correct line are subtle in comparison to the changes for Europe and Asia, notice that the line's attachment to the "20 million" mark at the last data point clearly puts it higher in 2020 than at any other point.

Example #2

Next, we're going to look at a non-science example. Notice that the answer choices here contain much more information than the ones in the previous question.

The length of the answer choices suggests that the graph may <u>not</u> be important, so we're going to start by just looking at the passage and the questions. We'll consider the graph later, if we need it.

Despite steady gains in readership and overall popularity, e-books are hardly positioned to replace print books. Rather, the two types of media complement each other, providing the same content in different forms. Print books and e-books each have unique attributes and serve distinct purposes, which vary by demographic and situational factors. Rearchers Yin Zhang and Sunali Kudva used data from the National Reading Habits Survey to examine book-format preferences in a variety of situations. Although the pair found that e-books are firmly established as an option due to ease of accessibility, they also **concluded that <u>print books</u> offer greater appeal to <u>most readers</u> in certain situations**.

1 Mark for Review

Which choice best describes data from the chart that supports Zhang and Kudva's finding?

Ⓐ A large majority of readers seeking immediate access to a book preferred the electronic option, with less than 25% of readers opting for print.

Ⓑ The percentage of readers who choose a format based on the variety of titles available was about 50% for both print books and e-books.

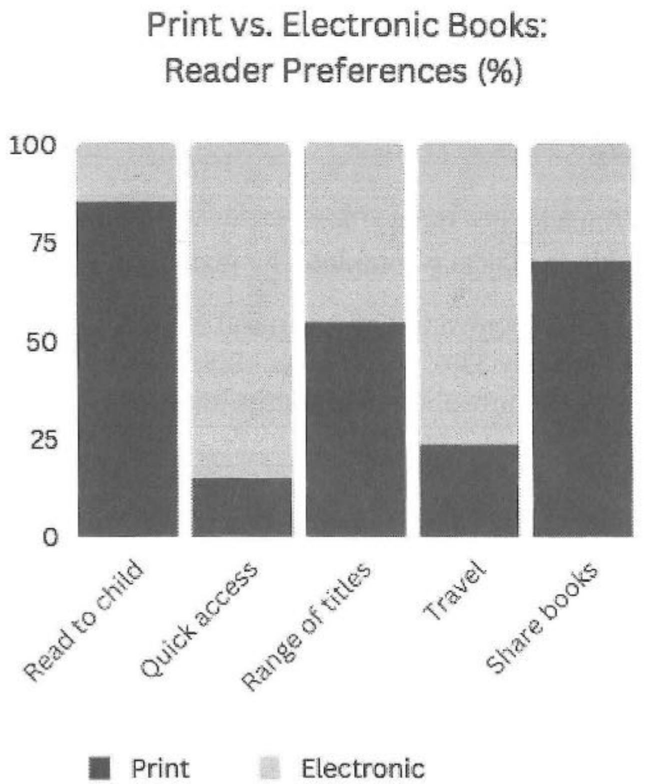
Ⓒ More than 75% of readers preferred print books for reading to a child, and more than 60% preferred print books for sharing with other readers.

Ⓓ When concerned about quick access or reading during travel, most readers preferred books in the same format.

The wording of the last sentence is key because it tells us what the researchers <u>concluded</u>: most readers—that is, **more than 50%**—prefer print books (dark gray bar) sometimes. Any answer that contradicts this idea or that has a different focus can be eliminated.

- (A) can be eliminated because the focus is on a situation in which readers prefer e-books.
- (B) doesn't fit because the focus is on a situation in which the numbers are roughly equal.
- (D) doesn't provide enough information to judge, although it is reasonable to assume that most readers who were traveling or seeking quick access to a book would want the electronic version.

(C) is a match, citing situations in which a clear majority of readers (75% and 60% respectively) prefer print books. It's almost certainly the right answer, but we're going to look at the graph to be safe.



Despite steady gains in readership and overall popularity, e-books are hardly positioned to take over print books. Rather, the two types of media complement each other, providing the same content in different forms. Print books and e-books each have unique attributes and serve distinct purposes, which vary by demographic and situational factors. Rearchers Yin Zhang and Sunali Kudva used data from the National Reading Habits Survey to examine book-format preferences in a variety of situations. Although the pair found that e-books are firmly established as an option due to ease of accessibility, they also **concluded that <u>print books</u> offer greater appeal to <u>most readers</u> in certain situations.**

1 Mark for Review

Which choice best describes data from the chart that supports Zhang and Kudva's finding?

Ⓐ A large majority of readers seeking immediate access to a book preferred the electronic option, with less than 25% of readers opting for print.

Ⓑ The percentage of readers who choose a format based on the variety of titles available was about 50% for both print books and e-books.

Ⓒ More than 75% of readers preferred print books for reading to a child, and more than 60% preferred print books for sharing with other readers.

Ⓓ When concerned about quick access or reading during travel, most readers preferred books in the same format.

Yes, the print-book bars (dark gray) for "Quick access" and "Travel" are both much smaller than the e-book bars (light gray) on top. That indicates that a much smaller percentage of readers prefer e-books in these situations. (D) can thus be conclusively eliminated, making (C) correct.

Example #3

Now back to a science passage. Once again, we're going to start by ignoring the graph entirely.

MRSA—*Methicillen-resistant Staphylococcus aureus*— is a type of bacteria that is frequently responsible for hospital-acquired infections and is known for being exceptionally resistant to common antibiotics. When a group of these medications were compared in terms of their ability to cure MRSA infections versus a variety of other infections, researchers **found that an antibiotic could demonstrate a high level of general effectiveness but that the same medication could be significantly less effective in combatting MRSA**. For example, _____

1 Mark for Review

Which choice most effectively uses data from the table to logically complete the text?

Ⓐ Erithromycin had a general effectiveness of nearly 32%, whereas its MRSA effectiveness was only about five points lower.

Ⓑ Whereas Mupirocin was around 17% less effective against MRSA than against general infections, the gap for Clindamicin was only about 14%.

Ⓒ Only Vancomycin was fully effective in combatting both general and MRSA bacteria.

Ⓓ Rifampicin showed a general effectiveness rate of more than 85%, whereas it was just over 60% effective against MRSA.

The key piece of information occurs in the sentence before the blank, which tells us that the same antibiotic can show very different effectiveness rates in general vs. against MRSA.

As a result, the correct answer must fulfill two conditions:

1) It must mention just one antibiotic.
2) It must include a statistic indicating a significant difference between general effectiveness and effectiveness against MRSA.

On that basis:

- We can eliminate (A) because 5% indicates a small gap.
- We can eliminate (B) because it compares two different medications.
- We can eliminate (C) because the effectiveness rates of Vancomycin were the same.

That leaves (D), which correctly cites an example of a single drug with a large difference (>25%) in effectiveness.

For the record, this is what the entire question looks like:

Antibiotic	General Effectiveness %	MRSA Effectiveness %
Erithromycin	31.94	26.92
Vancomycin	100	100
Mupirocin	90.28	73.08
Penicillin	5	0
Clindamicin	83.33	69.23
Rifampicin	86.11	61.54

MRSA—*Methicillen-resistant Staphylococcus aureus*— is a type of bacteria that is frequently responsible for hospital-acquired infections and is known for being exceptionally resistant to common antibiotics. When a group of these medications were compared in terms of their ability to cure MRSA infections versus a variety of other infections, researchers found that an antibiotic could demonstrate a high level of general effectiveness but that the same medication could be significantly less effective in combatting MRSA. For example, _____

1 Mark for Review

Which choice most effectively uses data from the table to logically complete the text?

Ⓐ Erithromycin had a general effectiveness of nearly 32%, whereas its MRSA effectiveness was only about five points lower.

Ⓑ Whereas Mupirocin was around 17% less effectiveness against MRSA than against general infections, the gap for Clindamicin was only about 14%.

Ⓒ Only Vancomycin was fully effective in combatting both general and MRSA bacteria.

Ⓓ Rifampicin showed a general effectiveness rate of more than 85%, whereas it was just over 60% effective against MRSA.

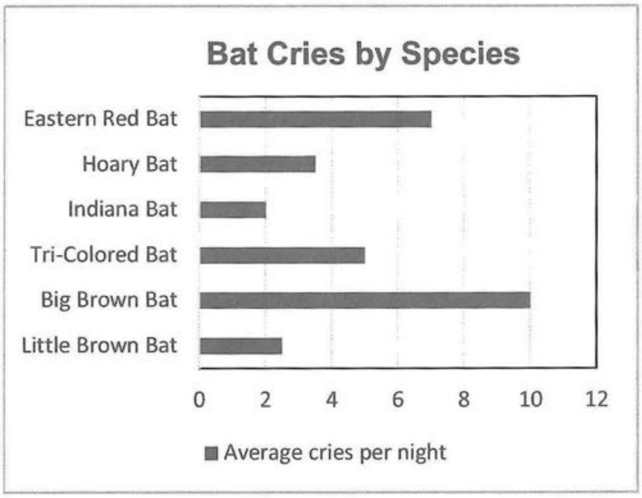
We could, of course, double-check (D) against the table, in which case we would see that the information is indeed cited accurately. But given that this answer is the only option that fulfills the requirements specified in the question, it must be correct regardless.

Note that all of the answer choices provide accurate descriptions of information in the graph, so if you looked only at the answers and the graph, you would have no way of determining the correct option.

Alternately, you might start by checking (A), notice that it was consistent with the graph, and choose that answer without bothering to check the others. **But to reiterate: an answer can accurately summarize information from the chart and still be wrong.**

If, on the other hand, you started with the passage and the question, you would avoid that trap completely and focus only on the relevant information from the start.

Exercise: Graphs and Charts



A student studying the presence of white-nose syndrome, a disease that has decimated bat populations across North America since 2007, examined data about the number of bats in a protected area. Because bats live in the dark and their cries are inaudible to people, biologists must use acoustic detectors to record their sounds and then analyze them to identify the species present in a given location. Observing that the highest number of cries came from big brown bats (*Eptesicus fuscus*), the student concluded that brown bat species were not affected by the disease.

1 Mark for Review

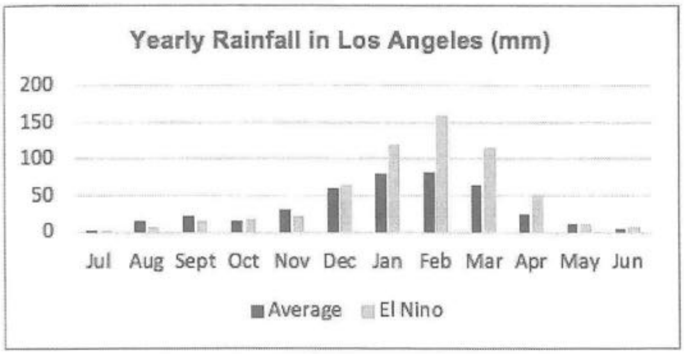
Which statement best describes data from the graph that would undermine the student's conclusion?

Ⓐ No bat species emitted more than 10 cries on average per night.

Ⓑ The number of cries from little brown bats was among the lowest of any species.

Ⓒ More cries were recorded for tri-colored bats than for hoary bats.

Ⓓ Eastern red bats were more vocally active than big brown bats.



El Niño is a climate pattern in which water in the Pacific Ocean near the equator becomes hotter than usual, affecting the atmosphere and weather around the world. Although El Niño climate conditions are unpredictable, they typically occur every few years and can change the weather in the United States, particularly in the southern states and in California. Climatologists have found that although El Niño years do not bring heavy rains every month, the difference in rainfall during the winter in El Niño years can be much greater than the difference during other seasons. Los Angeles, for example, receives around 15mm of rain on in May on average and during El Niño years, whereas ______

2 Mark for Review

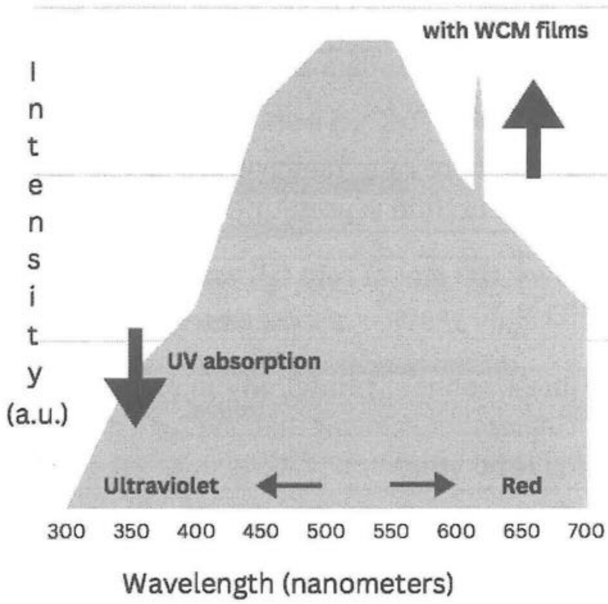
Which choice uses data from the graph to logically complete the text?

Ⓐ 5mm more rain than average falls in October when El Niño is present.

Ⓑ over 100 mm of rain falls in January during El Niño years.

Ⓒ the amount of rainfall in August and September is lower than average during El Niño years.

Ⓓ the amount of rainfall in February during El Niño years is 100 mm higher than average.



Plants use photosynthesis to create energy from visible light from the sun. In addition to visible light, however, sunlight contains ultraviolet (UV) light. Researchers at the University of Hokkaido wondered whether it would be possible to provide plants with additional visible light by employing a wavelength converting material (WCM) capable of transforming UV light into red light. They created a thin-film WCM coating and applied it to clear plastic sheets, which were placed next to Swiss chard plants. A control group used sheets without the coating. In the summer, when sun irradiation was strong, no significant difference was observed between the two groups. In winter, however, the plants with the WCM films were significantly taller and contained more biomass after 63 days as compared to the control group. The researchers concluded that this accelerated growth was caused by the increased supply of red light provided by the WCM films.

3 Mark for Review

Which choice uses data from the graph to support the researchers' conclusion?

Ⓐ Light that is less than 400 nanometers long had a much lower level of intensity than light that is more than 450 nanometers long.

Ⓑ Light ranging from 500 to 550 nanometers had the highest level of intensity.

Ⓒ The intensity of light waves more than 600 nanometers long that passed through the films increased to nearly peak levels.

Ⓓ UV absorption declined steeply in light waves that are less than 450 nanometers.

Study	No. of Participants	Sample
Erickson, 2010	39	Young adults, 18-28
Basek, 2011	20	Older adults, avg. 70.1
Vo, 2011	34	Young adults, 18-22
McGarry, 2013	7	Older adults, avg. 85
Kühn and Gallinat, 2014	62	Young adults, avg. 28.4
Szabo, 2015	56	Adults, avg. 36.8
Zhang, 2015	45	Adolescents, avg. 16.9
Takeushi, 2016	189	Children, 5-16
Takeushi, 2016	240	Children, avg. 11.1

Despite the sensationalist claims about the effects of video games on children's development that regularly appear in the media, a growing body of research purports to demonstrate that game players outperform non-gamers on a range of cognitive measures, and some studies suggest that the skills acquired through gaming can be transferred to real-world situations. However, these studies employ a variety of methodologies, criteria, and types of participants. Marc Palaus and colleagues at Oberta University in Spain conducted a review of 116 studies, aiming to better understand the relationship between gaming and cognitive development. While they concluded that it is possible to establish links between video games and skills involving attention, cognitive control, and visuospatial processing, they also observed that the lack of standardization could contribute to inconsistencies in the findings of similar studies.

4 Mark for Review

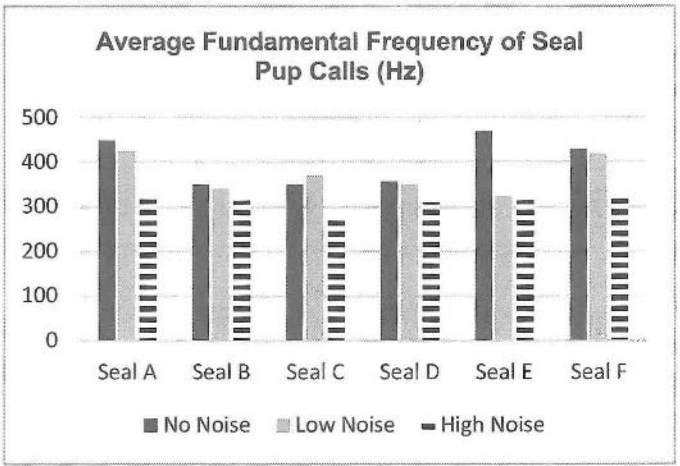
Which choice most effectively uses data from the graph to support the researchers' observation?

Ⓐ The studies run by Takeushi relied on larger sample sizes and involved younger participants than any of the other studies.

Ⓑ Few of the studies included participants under the age of 18.

Ⓒ Each study focused on participants in a particular age range and did not include members that were much older or younger.

Ⓓ The participants in the studies conducted by Erickson, Vo, and Kühn and Gallinat were all under the age of 30.



Seals are among the few mammals other than humans that are capable of learning new types of vocalizations. Whereas it is well established that adult harbor seals can acquire new vocal patterns, until recently this phenomenon had never been studied in pups. In 2021, researchers at the Max Planck Institute for Psycholinguistics and a group of colleagues conducted a study in which they played a series of pre-recorded sounds for a group of harbor seal pups ranging from one to three weeks old. They found that the baby seals were able to modify their vocalization patterns: overall, the pups lowered the fundamental frequency (FO) of their calls in response to increased noise. In some cases, the response with highly pronounced, with the same animal emitting vocalizations at much lower frequencies in high noise than in low noise.

5 Mark for Review

Which choice most effectively uses data from the table to support the researchers' finding?

Ⓐ Seal B emitted vocalizations of 330 Hz in low noise and 300 Hz in high noise.

Ⓑ The fundamental frequency of Seal A's vocalizations in high noise was 100 Hz lower than in low noise.

Ⓒ Whereas the fundamental frequency of Seal A's call was 125 Hz lower in high noise than in no noise, the frequency of Seal B's call dropped by less than 50 Hz.

Ⓓ In both low and high noise, the fundamental frequency of Seal E's call was around 150 Hz than it was without noise.

Answers: Graphs and Charts

1. B

Although it may look complicated, this question is actually quite straightforward. What is the student's conclusion? That *brown bat species were not affected by [white-nose syndrome]* because the highest average number of cries recorded came from big brown bats. If you look at the graph, you can see that there are actually two brown bat species listed, and the number of cries for the other species (little brown bat) is at the very low end—a finding that would imply some brown bats are in fact affected. That is what (B) says, so it is correct. All of the other answers are off-topic.

2. D

Although the passage is fairly long, the key information—as usual—appears at the end. You cannot just read the last sentence, however: it includes the transition *for example,* which means that you need to back up and find the idea that the sentence is illustrating. You'll find it in the previous sentence, which states that although El Niño does not result in consistently heavy rains throughout the year, *the difference in rainfall during the winter in* El Niño *years can be much greater than the difference during other seasons.* Los Angeles is then cited as an example of a city where this is the case. The last sentence indicates that average and El Niño rainfall are the same in May (in a season other than winter), and the word *whereas* before the blank indicates that the remainder of the sentence must describe the opposite—a winter month in which the difference between average and El Niño rainfall is large. The correct answer must therefore include a comparison between *El Niño* rainfall and average rainfall.

(B) does not include a comparison, so this answer can be eliminated immediately. Be careful with (A): this answer does mention a month in which El Niño is associated with more rain than average, but 5mm is a very small amount, and the correct answer must involve an amount that is *much greater.* (C) does not fit because the correct answer must involve months in which El Niño rainfall is higher, and there is also no information about the size of the differences. (D) correctly cites an example of a winter month (February) in which El Niño is associated with rainfall that is much higher than the average (150 mm).

3. C

Don't be intimidated by the seeming complexity of the graph—this question is much easier than it looks. What is the researchers' conclusion? You'll find it in the last sentence: the accelerated growth of the plants with WCM films *was caused by the increased supply of red light.* The correct answer must support the idea that the films increased the red-light supply.

Shortcut: (D) is the only answer that mentions the films, so start by checking it. Indeed, the fact that the light that passed through the films increased to "peak levels" of intensity in the "red" area of the graph directly illustrates the idea that the films caused the plants to receive more red light. (D) is thus correct. The other answers include information about light wavelength, intensity, and absorption but do not explicitly mention the effect of the films.

4. A

What is the researchers' observation? That *the lack of standardization [among studies] could contribute to inconsistencies in the findings of similar studies.* The correct answer must therefore mention a factor that varies from study to study and include a comparison of some type. The words *Each study* in (C) and *all* in (D) indicate that these answers are discussing factors that remain the same across studies, so both choices can be eliminated. (B) is off-topic: even though the studies involve video games, the relative lack of children among the participants is not the issue indicated by the passage. Only (A) mentions an inconsistency that illustrates the researchers' observation: the studies run by one of the participants differed noticeably from the other studies in that they involved far more participants. In other words, it shows that the studies were not standardized.

5. B

What is the researchers' finding? It appears in the last sentence: the seal pups changed their vocalizations in response to the sounds, and in some cases, <u>the same</u> animal might produce vocalizations with <u>much lower</u> fundamental frequencies in high noise than in low noise. The correct answer must therefore focus on one animal only and indicate a large gap in fundamental frequency between low and high noise. (C) makes a comparison between two seals, so it can be eliminated automatically. Be careful with (A): it does compare the frequencies in low and high noise, but the gap is fairly small (30 Hz). (D) is likewise incorrect because the correct answer must cite a large difference in the responses to the two noise levels, and here it is very similar. Only (B) offers an example of a seal whose vocalizations showed a large gap (100 Hz) between low and high noise.

10 Paired Passages

While the digital SAT will consist primarily of single short passages accompanied by one question, it will also contain a set of short paired passages ("Text 1" and "Text 2") that present alternate viewpoints or information about the same topic. These passages will be accompanied by a question asking about the relationship between them.

In previous versions of the exam, and mostly likely in this version as well, the most common relationship between the passages was **disagreement**. However, the two authors—or figures mentioned in one or both or the passages—may **also agree** on certain points. As a result, you must be able to distinguish between aspects of the topic on which the two opinions overlap and those where they differ. **Correct answers can involve areas of both agreement and disagreement.**

In addition, answers to paired-passage questions may depend on an easily overlooked detail in one or both passages. Sometimes that detail will be located in a key place (first sentence, last sentence, close to a major transition), but sometimes it will not. Because you may not remember the key information, you should always return to the passages as needed. You should also make sure not to eliminate any answer you're uncertain about until you have confirmed that it is incorrect.

If you are a strong reader who handles this type of writing well, you can probably approach paired passages much as you approach the other type of passages. However, **if you are not aiming for a very high score and struggle disproportionately with this question type, you should plan to skip it initially and only return after you have answered everything that you can answer easily.**

Even though both passages will be very short, they may still be dense and discuss unfamiliar subjects. In order to avoid confusion, **you should aim to deal with the smallest amount of information possible at any given time**. The more work you do in terms of determining arguments upfront, the less work you'll need to do later. You can move very quickly through the steps outlined on the following page, but for maximum effectiveness, you should not skip any of them.

How to answer paired-passage questions:

1) Read Text 1: identify main point.

2) Read Text 2: identify main point.

3) Write the relationship.

4) Answer the question in your own words.

5) Look at the choices, and pick the one closest to your answer.

It is important to **determine the relationship** between the passages upfront because the question will always test your understanding of that connection, typically asking you what the author of one passage would "say about" or "respond to" an idea in the other.

Once you have this information, you've essentially answered the question before you've even looked at it. If the authors of the two passages disagree only, the correct answer will be negative. You can thus eliminate positive and neutral options, which can often be identified from the first few words of the answer. Furthermore, when answers include more detailed information about the passages, you may sometimes be able to use the main points themselves.

You can also use your knowledge of the test's structure to help you make educated guesses. For example, take a look at the question below. We don't even need a set of passages here—our only concern is how to use the framework of the test to predict the most probable answer.

1 Mark for Review

How would Chang (Text 2) most likely respond to the researchers' theory about sustained stress in Text 1?

Ⓐ It is not convincing as an explanation, despite findings that may seem to support it.

Ⓑ It is generally accurate but contains some questionable elements.

Ⓒ It can only be applied in certain situations.

Ⓓ It may seem reasonable, but the researchers' results do not support it.

Since passages generally disagree, we can assume that the correct answer will be negative. (B) is positive ("generally accurate"), so it's probably wrong. (C) is negative, but it contains the extreme word "only," so we'll eliminate it as well. Between (A) and (D), (D) is more negative, so we're going to give a slight edge to that option. Besides, when there's a mismatch between theory and practice, it's usually because the theory sounds good but doesn't work in practice—not the other way around, as (A) indicates. At this point, we could go back to Text 2 and check for a mention of findings that would support (D). If that didn't work, we would go to (A), then (C).

Now let's look at a full-length example.

Text 1

Our food now travels an average of 1,500 miles before ending up on our plates. This globalization of the food supply has serious consequences for the environment, our health, our communities and our tastebuds. Much of the food grown in the breadbasket surrounding us must be shipped across the country to distribution centers before it makes its way back to our supermarket shelves. Because uncounted costs of this long-distance journey (air pollution and global warming, the ecological costs of large-scale monoculture, the loss of family farms and local community dollars) are not paid for at the checkout counter, many of us do not think about them at all.

Text 2

Just how much carbon dioxide is emitted by transporting food from farm to fork? Pierre Desrochers and Hiroko Shimizu cite a comprehensive study done by the United Kingdom's Department of Environment, Food and Rural Affairs (DEFRA) which reported that 82 percent of food miles were generated within the U.K. Consumer shopping trips accounted for 48 percent and trucking for 31 percent of British food miles. Air freight amounted to less than 1 percent of food miles. In total, food transportation accounted for only 1.8 percent of Britain's carbon dioxide emissions.

1 Mark for Review

Based on the texts, how would Desrochers and Shimizu (Text 2) most likely describe the view presented in Text 1?

Ⓐ It is strongly supported by data compiled by DEFRA.

Ⓑ It overstates the effects of transporting food on the environment.

Ⓒ It appears justified by preliminary findings but has not yet definitively proven.

Ⓓ It is highly implausible because most consumers do not consider the source of their food.

Let's start by breaking things down.

Topic:

Transporting food long distances to sell.

1) Main Point, Text 1:

Transporting food long distances harms environment.

2) Main Point, Text 2:

Transporting food long distances doesn't really harm environment.

3) Relationship:

Disagree.

4) Answers:

(A) Positive - eliminate it.

(B) Negative - keep it.

(C) Positive - eliminate it.

(D) Negative - keep it.

Now, be careful with (D). You might remember something about people not thinking about where their food comes from, but that's in Text 1. If you go back to Text 2, you'll find it says nothing about that idea.

Text 2 does, however, state that *food transportation accounted for only 1.8 percent of Britain's carbon dioxide emissions,* which stands in contrast to the doom-and-gloom environmental scenario presented in the first passage. Based on that statement, you can assume that Desrochers and Shimizu would consider the claim in Text 1 to be somewhat exaggerated, i.e., "overstated." That makes (B) correct.

Exercise: Paired Passages

Text 1

By investigating interactions between tree species, scientists have found that trees leverage similarities and differences in their microbial "makeup" to recognize other trees of their own species, and that they preferentially share nutrients with them through their mycorrhizal network—the systems of roots and fungi that connect them. For example, Douglas Fir trees growing in the same plot have been shown to share more carbon among them than with trees of other species.

Text 2

The notion that trees send out resources to strengthen a community composed of members of their species is unlikely because groups that cooperate would need to win out over groups made up of competing individuals. According to plant ecologist Kathryn Flinn, while trees can sometimes facilitate each other's growth, a forest does not function like a single organism: it includes a vast array of species with a constantly shifting variety of interactions, both cooperative and competitive.

1 Mark for Review

Based on the texts, what would Kathryn Flinn most likely say about the "Douglas Fir trees" in Text 1?

Ⓐ Their mycorrhizal network is not fully understood.

Ⓑ They function as if they were a single organism.

Ⓒ They are also likely to compete among themselves for some resources.

Ⓓ The amount of carbon they share will vary according to environmental conditions.

Text 1

In recent years, there has been an explosion of scientific research revealing precisely how positive feelings are beneficial. We know that they motivate people to pursue important goals and overcome obstacles, offer protective benefits against the effects of stress, improve our social connectedness, and even ward off illness. The science of happiness has spawned a small industry of motivational speakers and research enterprises. Clearly, happiness is popular.

Text 2

Happiness, it turns out, has a cost when experienced too intensely. For instance, we often are told that happiness can open up our minds to foster more creative thinking and help us tackle problems or puzzles. This is the case when we experience moderate levels of happiness. But according to Mark Alan Davis's 2008 analysis of the relationship between mood and creativity, when people experience intense and perhaps overwhelming amounts of happiness, they no longer experience the same creativity boost. What's more, psychologist Barbara Fredrickson has found that too much positive emotion—and too little negative emotion—makes people inflexible in the face of new challenges.

2 Mark for Review

Based on the texts, what would Mark Alan Davis most likely respond to what "we know" in Text 1?

Ⓐ By emphasizing the connection between creativity and negative emotions

Ⓑ By acknowledging the benefits of positivity in moderation but cautioning against it in excess

Ⓒ By questioning the motives of the participants in the happiness industry

Ⓓ By challenging the connection between positive feelings and personal fulfillment

Text 1

Until recently, the concrete psychological effects of fiction on individuals and society were largely a matter of speculation. However, research in psychology is beginning to provide answers about how fiction can expand our moral imaginations. For example, a series of studies conducted by Keith Oatley, Maja Djikic, and Raymond Mar found that fiction measurably improves people's ability to guess others' mental states by looking at only their eyes. They interpreted this finding as evidence for the idea that fiction allows people to connect with something larger than themselves.

Text 2

An empirical approach to the question of whether fiction improves empathy was taken by David Kidd and Emanuele Castano, who conducted five experiments in which participants read fictional excerpts and then responded to images of facial expressions. The results showed that the participants had improved their theory of mind (ToM), or their ability to infer the thoughts and emotions of others. As Kidd points out, however, highly developed ToM does not always translate into more ethical behavior: the ability to manipulate someone, for instance, also requires a heightened understanding of other people's emotions.

3 Mark for Review

Based on the texts, how would Kidd and Castano most likely respond to Oatley, Djikic, and Mar in Text 1?

Ⓐ By acknowledging the importance of connecting with others

Ⓑ By conceding that fiction can allow people to transcend their everyday lives

Ⓒ By pointing out that empathy can have negative as well as positive effects

Ⓓ By emphasizing that individuals with high ToM may sometimes prefer non-fiction

Text 1

On May 21, 2019, midsize black holes were detected for the first time when the U.S.-based Laser Interferometer Gravitational-Wave Observatory (LIGO) and its European counterpart Virgo captured a tremor from a pair of black holes merging deep in space. Priyamvada Natarajan, an astrophysicist who has long worked on black-hole growth models, believes that black holes this size are born in nuclear star clusters, dense collections of stars found near galactic centers. These holes sweep through the cluster, adding gas and dust, until they settle at a single location and cease to expand.

Text 2

Imre Bartos and other researchers working on "hierarchical merger" models, in which black holes grow by eating one another, focus on one major data point in the LIGO/Virgo findings. The angular momentum, or "spin," of a black hole ranges from 0 to 1. When two black holes of similar size combine, the resulting black hole usually has a spin of around 0.7. Significantly, the two black holes involved in the merger recorded by LIGO and Virgo had 0.69 and 0.73 respectively, suggesting that they both might have formed in previous mergers.

4 Mark for Review

Based on the texts, what would Imre Bartos most likely say about Priyamvada Natarajan's belief in Text 1?

Ⓐ It underestimates midsize black holes' spin.

Ⓑ It misstates the time when the merger occurred.

Ⓒ It relies too heavily on data from LIGO/VIGO.

Ⓓ It overlooks the significance a crucial statistic.

Answers: Paired Passages

1. C

What does Text 1 say about Douglas Fir trees? *When they grow in the same plot, they share more carbon among them than with trees of other species.*

Next, what is Kathryn Flinn's position (Text 2)? She asserts that while trees *can sometimes help facilitate each other's growth,* relationships among organisms in forest are *constantly shifting* and include both cooperation and competition.

Logically, how would Flinn respond to the observation that Douglas firs seem to "prefer" sharing carbon with members of their own species? Given her perspective that species both cooperate and compete, she would probably respond to an example of cooperation by suggesting that the Douglas firs also compete at times. And that is exactly what (C) says.

(A) and (D) are irrelevant to Flinn's discussion, and (B) is directly contradicted by her assertion that *a forest does not function like a single organism.*

2. B

What does Text 1 state that "we know"? If you scan the passage for these words, you'll find them at the beginning of the second sentence, which tells us that positive feelings are associated with a host of happy outcomes (motivation to achieve goals, overcome obstacles, protect against stress, etc.).

Next, what position does Mark Alan Davis (Text 2) hold? That *when people experience intense and perhaps overwhelming amounts of happiness, they no longer experience the same creativity boost.* In other words, too much positivity is not a good thing.

Logically, how would Davis respond to the example from Text 1? Presumably by reiterating his position that while moderate amounts of happiness are helpful, too much of it is not, i.e., "by acknowledging [its] benefits but cautioning against it in excess." (B) is thus correct.

(A) is incorrect because Davis's position on negative emotions is never mentioned. (C) is beyond the scope of what can be inferred from Davis's position.

Be careful with (D): Davis is only mentioned in relation to creativity, not "personal fulfillment," and he does not deny the beneficial effects of happiness—he only indicates that too much of it can have negative effects.

3. C

What is the position of Oatley, Djikic, and Mar in Text 1? Fiction *improves people's ability to guess others' mental states* and improves their ability to empathize.

What is the position of Kidd and Castano in Text 2? They agree that fiction improves people's understanding of others' mindsets but point out that these skills can be used in negative ways.

Logically, how would Kidd and Castano respond to the position in Text 1? That position only emphasizes the positive aspect reading fiction, so they would presumably point out the negative one. And that is what (C) says.

(A) is outside the scope of the passages, which focus on the effects of fictions rather than relationships in general. (B) is incorrect because Kidd and Castano only discuss the effects of fiction on theory of mind, not on "transcend[ing] everyday lives." (D) is entirely off-topic; neither passage discusses or implies anything about the effects of reading non-fiction.

4. D

What is Priyamvada Natarajan's belief in Text 1? That *black holes this size* (i.e., of medium size) come from nuclear star clusters and sweep through them gathering gas and dust.

What is Imre Bartos's position (Text 2)? *Black holes grow by eating one another,* a view that is based on an analysis of black holes' spin— a specific statistic that falls within a narrow range.

Logically, how would Bartos respond to Natarajan? Because it is not mentioned that Natarajan takes "spin" into account, Bartos would presumably focus on the absence of that figure, i.e., "a crucial statistic." That makes (D) correct.

(A) is incorrect because Natarajan's model does not include spin at all. (B) is incorrect because Bartos does mention when the merger occurred. (C) is incorrect because Bartos's model, not Natarajan's, is mentioned as relying on specific data from LIGO/VIGO.

Reprints and Permissions

Adee, Sally. From "Roughnecks in Space: Moon Mining in Science Fiction," *New Scientist*, 27 April 2012. http://www.newscientist.com/gallery/moon-mining

Anft, Michael. "Solving the Mystery of Death Valley's Walking Rocks," *Johns Hopkins Magazine*, 6/1/11. Reprinted by permission of *Johns Hopkins Magazine*. http://archive.magazine.jhu.edu/2011/06/solving-the-mystery-of-death-valley's-walking-rocks/

Austen, Jane. *Northanger Abbey*. Originally published 1803. Excerpted from Chapter 1 through Project Gutenberg, http://www.gutenberg.org/files/121/121-h/121-h.htm

Ball, Philip. From "The Trouble with Scientists," *Nautilus*, 5/14/15. http://nautil.us/issue/24/error/the-trouble-with-scientists

Bibb, Eloise. "Gerarda," 1895. https://www.poetrynook.com/poem/gerarda

Bordal, Laura Torres; Jadoul, Yannick; Rasilo, Heikki; Salazar Casals, Anna; Ravignani, Andrea. "Vocal Plasticity in Harbour Seal Pups,"Comparative Bioacoustics Group, Max Planck Institute for Psycholinguistics, Nijmegen, The Netherlands. https://doi.org/10.1101/2021.02.10.430617

Brauer, Wiebke. Adapted from "The Miracle of Space," *Smart Magazine*, 10/31/14. Reprinted by permission of *Smart Magazine. http://smart-magazine.com/space/the-miracle-of-space/*

Brontë, Emily. *Wuthering Heights*. Originally published 1847. Excerpted from Chapter XX, http://www.online-literature.com/bronte/wuthering/20/

Chopin, Kate. *The Awakening*, originally published 1899, Chapter 6. Accessed through Project Gutenberg, https://www.gutenberg.org/files/160/160-h/160-h.htm

Dickinson, Emily. "Beclouded." Originally published 1896. https://etc.usf.edu/lit2go/114/the-poems-of-emily-dickinson-series-one/2632/nature-poem-29-beclouded/

Dunbar, Paul Laurence. "Frederick Douglass," 1895. https://poets.org/poem/frederick-douglass

Freedman, David. Adapted from "The Truth About Genetically Modified Food," Scientificamerican.com (Nature, Inc.), September 1, 2013. https://www.scientificamerican.com/article/the-truth-about-genetically-modified-food/

Gholipour, Bahar. Adapted from "Sleep Shrinks the Brain's Synapses to Make Room for New Learning," Scientificamerican.com (Nature, Inc.), May 1, 2017. https://www.scientificamerican.com/article/sleep-shrinks-the-brain-rsquo-s-synapses-to-make-room-for-new-learning/

Grimké, Charlotte Forten. "Wordsworth," https://www.beltwaypoetry.com/grimke-charlotte-forten/

Gross, Daniel. From "Will We Ever Be Able to Make Traffic Disappear?" Smithsonian.com, 5/7/2015. http://www.smithsonianmag.com/innovation/will-we-ever-be-able-to-make-traffic-disappear-180955164/

Hamlin, Kimberly. "Monumental Women," *Humanities*, 8/10/2020. https://www.neh.gov/article/monumental-women

Heard, Josephine. "On receiving Tennyson's Poems from Mrs. M, H. Dunton, of Brattleboro, Vt," 1890. https://scalar.lehigh.edu/african-american-poetry-a-digital-anthology/josephine-heard-morning-glories-full-text-1

Hokkaido University, graphic: plants and red light. https://www.global.hokudai.ac.jp/blog/uv-to-red-light-converting-films-accelerate-plant-growth/

Johnson, Georgia Douglas. "Youth," 1922. https://poets.org/poem/youth-2

Kaufman, Les. Adapted from "One Fish, Two Fish, Red Fish, Blue Fish," *National Geographic*, 2005.

Kelemen, Peter B. From "The Origin of the Ocean Floor," *Scientific American*, February 2009. **Reproduced with permission, © 2009, Scientific American, Inc. All rights reserved.**
http://www.scientificamerican.com/article/the-origin-of-the-ocean-floor/

Koerth-Baker, Maggie: From "The Power of Positive Thinking, Truth or Myth?" *Live Science*, 8/29/08.
http://www.livescience.com/2814-power-positive-thinking-truth-myth.html

Killam, Daniel. El Niño graph, adapted from "California Getting Wetter to the North, Drier to the South: Natural Variability or Climate Change?" 10.3390/cli2030168

Lehrer, Jonah. Adapted from "Under Pressure: The Search for Stress Vaccine," *Wired*, 7/28/10, 2:00 p.m.
http://www.wired.com/2010/07/ff_stress_cure/

Mills, Mark. From "Every Breath You Take," *City Journal*, 7/2/13. http://www.cityjournals.org/2013/bc0702mm.html

Nuwer, Rachel. "What is a Species?" Smithsonian.com, 11/6/13. http://www.smithsonianmag.com/science-nature/what-is-a-species-insight-from-dolphins-and-humans-180947580/?no-ist

Orwell, George. Adapted from "Keep the Apidistra Flying," originally published 1936. Accessed from Project Gutenberg, http://gutenberg.net.au/ebooks02/0200021.txt

Palaus, Marc. Chart data adapted from "Neural Basis of Video Gaming: A Systematic Review." *Frontiers of Human Neuroscience*, vol. 11, 2017. https://doi.org/10.3389/fnhum.2017.00248

Panko, Ben. "What Does it Mean to Be a Species? Genetics is Changing the Answer," Smithsonian.com, May 19, 2017. http://www.smithsonianmag.com/science-nature/what-does-it-mean-be-species-genetics-changing-answer-180963380/

"Raining Pebbles: Rocky Exoplanet Has Bizarre Atmosphere, Simulation Suggests," Washington University in St. Louis, 10/1/09. https://www.sciencedaily.com/releases/2009/09/090930165038.htm

Rupp, Rebecca. From "Surviving the Sneaky Psychology of Supermarkets," *National Geographic*, 6/15/15.
http://theplate.nationalgeographic.com/2015/06/15/surviving-the-sneaky-psychology-of-supermarkets/

Sainani, Kristin. Adapted from "What, Me Worry?" *Stanford Magazine*, May/June 2014.
https://alumni.stanford.edu/get/page/magazine/article/?article_id=70134. Reprinted by permission of the author.

Schwartz, Barry. "More Isn't Always Better," *Harvard Business Review*, June 2006. Reprinted by permission of Harvard Business Publishing. https://hbr.org/2006/06/more-isnt-always-better

"Scientists Discover Salty Aquifer, Previously Unknown Microbial Habitat Under Antarctica," © 2015, Dartmouth College. http://www.sciencenewsline.com/articles/2015042815560014.html

Sullivan, John Jeremiah. "One of Us," *Lapham's Quarterly*, Spring 2013.
http://www.laphamsquarterly.org/animals/one-us

Excerpt from THE BONESETTER'S DAUGHTER by **Amy Tan**, copyright 2001 by Amy Tan. Used by permission of G.P. Putnam's Sons, an imprint of Penguin Publishing Group, a division of Penguin Random House LLC. All rights reserved.

Taylor, Astra. "A Small World After All," *Bookforum*, June/July/August 2013. http://www.bookforum.com/inprint/020_02/11685

Thompson, Helen. "Yawning Spreads Like a Plague in Wolves," Smithsonian.com, 8/27/14, http://www.smithsonianmag.com/science-nature/yawning-spread-plague-wolves-180952484/

Tregaskis, Sharon. "The Buzz: What Bees Tell Us About Global Climate Change, *Johns Hopkins Magazine*. 6/2/2010. Reprinted by permission by *Johns Hopkins Magazine*. http://archive.magazine.jhu.edu/2010/06/the-buzz-what-bees-tell-us-about-global-climate-change/

Updike, John. Adapted from "Big Bad Bizarre Dinosaurs {Extreme Dinosaurs}, *National Geographic*, December 2007.

Wayman, Erin. "The First Americans," Smithsonian.com, 11/28/11. http://www.smithsonianmag.com/science-nature/the-first-americans-889302/

Wharton, Edith. *Summer*, originally published 1917. Excerpted from Chapter 1 through Project Gutenberg, http://www.gutenberg.org/files/166/166-h/166-h.htm

Wilde, Oscar. *The Importance of Being Earnest*, originally published 1895. Excerpted from Act I, Scene I through Project Gutenberg, https://www.gutenberg.org/files/844/844-h/844-h.htm

"Why Eat Locally?" *The Regal Vegan*, http://www.regalvegan.com/site/local-yokels/

Zhang, Y. and Kudva, S. "E-books versus print books: Readers' choices and preferences across contexts," Journal of the Association for Information Science & Technology, Association for Information Science & Technology, vol. 65(8), pages 1695-1706, August. Graphic information based on Pew Research Statistics.

ABOUT THE AUTHOR

Erica Meltzer earned her B.A., magna cum laude, from Wellesley College and spent more than a decade tutoring privately in Boston and New York City, as well as nationally and internationally via Skype. Her experience working with students from a wide range of educational backgrounds and virtually every score level, from the third percentile to the 99th, gave her unique insight into the types of stumbling blocks students often encounter when preparing for standardized reading and writing tests.

She was inspired to begin writing her own test-prep materials in 2007, after visiting a local bookstore in search of additional practice questions for an SAT Writing student. Unable to find material that replicated the contents of the exam with sufficient accuracy, she decided to write her own. What started as a handful of exercises jotted down on a piece of paper became the basis for her first book, the original Ultimate Guide to SAT Grammar, published in 2011. Since that time, she has authored guides for SAT reading and vocabulary, as well as verbal guides for the ACT®, GRE®, and GMAT®. Her books have sold more than 300,000 copies and are used around the world. She lives in New York City, and you can visit her online at www.thecriticalreader.com.

The Critical Reader: The Complete Guide to SAT® Reading is designed to clearly and systematically demystify what is often considered the most challenging part of the exam.

Includes:

- A chapter-by-chapter breakdown of question types, with in-depth explanations and numerous examples demonstrating how to work through each category.
- Techniques for understanding challenging passages and identifying key information quickly and efficiently.
- Strategies for simplifying and answering graphic-based questions.
- A list of high-frequency vocabulary words grouped by category, plus strategies for determining the meanings of words used in unfamiliar ways.

Whether you are seeking a comprehensive review or need to focus on just a few key areas, this book teaches you the skills and strategies you need to master the Reading portion of the digital SAT with confidence.

In 2007, Erica Meltzer answered a Craigslist ad for an SAT tutor and unexpectedly found herself in the high-pressure world of Manhattan test prep. Today, she is the author of guides for the SAT®, ACT®, GRE®, GMAT® and IELTS®. Her books are used around the world, and her work has been cited in *Teen Vogue*, *The College Solution*, and *CBS MarketWatch*. She lives in New York City, and you can visit her online at www.thecriticalreader.com.

THE CRITICAL READER

ISBN 9798987383513

90000

9 798987 383513